Frozen Progress: The Poverty Cost of an Extreme Winter in Mongolia

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Abstract

In the winter of 2023/24, Mongolia lost one in eight of its livestock, more than eight million head, to a dzud: an extreme winter of deep cold, heavy snow, and ice-locked pasture. Yet national poverty fell steeply between 2022 and 2024. Cold extremes are largely absent from the disaster-poverty evidence; exploiting spatial variation in winter severity in the first nationally representative household panel to bracket a major dzud, we provide, to our knowledge, the first quasi-experimental estimate of their monetary poverty cost. Poverty in the most affected districts would otherwise have fallen roughly 15 percentage points further. Herders bore the cost through asset destruction: herds in the worst-hit districts ended about a third smaller, and own-produced food went with the herd. Non-herders were spared the asset shock but not the food shock: food insecurity rose by roughly four-fifths (8 percentage points on a 10 percent base). In both groups the burden fell on poorer households. As climate extremes intensify, cold shocks in pastoral economies can reverse poverty gains unseen in aggregate statistics.

Keywords: Poverty; Food insecurity; Cold extreme; Climate shocks; Pastoralism

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1 Introduction

In the winter of 2023/24, Mongolia lost more than eight million head of livestock, one in every eight in the country, to a dzud: an extreme winter of deep cold, heavy snowfall, and ice-locked pasture (UNDP, 2025). The aggregate record shows no trace of it. Over the two years that bracket that winter, a mining-led boom drove the national poverty rate from 27 to 16 percent, and judged by these statistics the country's worst winter disaster in more than a decade cost nothing.[1] This paper asks what the national decline concealed: whether the cold extreme increased poverty, through what channels, and for whom.

The disaster-poverty literature has a blind spot where this event sits. Weather extremes are becoming more frequent and more intense (Seneviratne et al., 2021); disasters already push an estimated 26 million people below the poverty line each year (Hallegatte et al., 2020).[2] A substantial body of work traces how such shocks reach household welfare through asset loss, private coping, and persistent livelihood effects (Sawada and Takasaki, 2017), yet cold extremes are largely absent from this evidence (Spencer and Strobl, 2025; Hallegatte et al., 2020; Karim and Noy, 2016; Hallegatte et al., 2016).[3] Where cold has been studied, the outcome is mortality or health rather than household welfare (Deschênes and Moretti, 2009; Barreca et al., 2016). The dzud literature itself documents lasting non-monetary scars, child stunting, interrupted schooling, slow asset recovery, and out-migration, while the first-round effects on poverty, consumption, and income remain largely unmeasured (Groppo and Kraehnert, 2016, 2017; Lehmann-Uschner and Kraehnert, 2018; Roeckert and Kraehnert, 2022).

The hazard itself is anything but marginal. Three catastrophic dzuds in twenty-five years together killed close to thirty million head of livestock (UNDP, 2025).[4] The compound summer-drought-and-severe-winter conditions that produce a dzud have co-

[1] The national headcount fell from 27.1 to 15.8 percent. In the estimation sample (provincial Mongolia, excluding the capital, Ulaanbaatar), the population-weighted headcount fell from 31.5 to 20.7 percent; within it, poverty in the worst-hit (treated) districts barely moved, from 23.4 to 22.7 percent, against a fall from 33.5 to 20.2 percent in control districts (Figure 2; authors' calculations from the HSES panel).

[2] Under adverse scenarios, climate change could add more than 100 million people to poverty by 2030 (Hallegatte et al., 2016).

[3] None of the four yields a household-welfare estimate from a cold-zone or pastoral setting; the most recent excludes "winter storms, cold spells" by design (Spencer and Strobl, 2025, p. 952).

[4] The consecutive dzud winters of 1999–2002 killed 11.2 million head of livestock; the single winter of 2009/10 killed 10.3 million, 23 percent of the national herd; and the 2023/24 event studied here killed 8.1 million (UNDP, 2025).

occurred more often since the turn of the century (Haraguchi et al., 2022). Nor is the exposure Mongolian alone: snow disasters kill yaks and sheep across the Qinghai–Tibetan Plateau (Li et al., 2018), cold waves kill alpaca herds on the Andean altiplano (Bazo et al., 2021), harsh winters push Kyrgyz livestock farmers into herd-eroding distress sales (Sultakeev and Petrick, 2025), and worldwide an estimated 200 to 500 million pastoralists hold their wealth in livestock that a single weather extreme can destroy (FAO, 2017).

Measuring this gap matters because current poverty, through income and consumption, is the first-round effect through which the longer-run scars documented above must run, and the margin on which disaster response can still act in real time. These monetary outcomes are also the ones most central to climate-poverty analysis (Hallegatte and Rozenberg, 2017) and to the design of anticipatory social protection (Bowen et al., 2020; Premand and Stoeffler, 2022). How large that first-round loss is, and how well households can buffer it, depend on whether the shock destroys earning capacity. Hurricane Katrina destroyed homes and displaced households on a massive scale, yet longer-run evidence finds its victims' income losses were temporary: the labor that generated their incomes survived the storm and moved with them (Deryugina et al., 2018).[5] When the earning asset itself dies, the first-round loss is large and poorly buffered.

For a herder, the question of what survives has a harsher answer: the earning capacity itself is the herd, and it is precisely what the winter kills. When the animals die, income, food from own production, collateral, and the breeding stock needed to rebuild fall together. Options remain: a herder can sell labor, move to a mining town, or leave the countryside altogether. Yet each of these replaces the livelihood rather than restores it, and rebuilding the herd is paced by the biology of the animals rather than by the household's effort.

Mongolia is the natural place to measure the first-round cost of such a shock: outside the capital, roughly a third of households herd as their primary livelihood, and their main productive assets are livestock standing directly in the path of winter risk. The dzud is also an unusually clean case: it consumes the herd while leaving housing, infrastructure, and the wider local economy comparatively untouched, so the measured cost is attributable to the asset loss itself. We study the 2023/24 episode with the first nationally representative household panel to bracket a major dzud, designed with the

[5] Displaced residents tracked through individual tax records were persistently displaced but largely recovered their incomes within a few years.

National Statistical Office of Mongolia as a subsample of the 2022 and 2024 Household Socio-Economic Survey (HSES).[6] Because the survey records income by detailed source, the panel supports a direct test of the mechanism: we can measure how much of the welfare loss runs through livestock income rather than infer it from herder status alone.

Identification exploits spatial variation in winter severity. We define a soum, Mongolia's district-level administrative unit, as exposed if its winter temperature or snow-cover anomaly exceeded the 85th percentile of its own historical distribution, the standard treatment definition in the dzud literature (Groppo and Kraehnert, 2016, 2017), and we exclude the capital, which has essentially no pastoral sector and belongs in neither the treated nor the control group. A difference-in-differences (DD) design compares exposed and less-exposed soums before and after the dzud, and a triple difference (TD) isolates the herder-specific asset loss by comparing herders and non-herders within the same soum. Neither the 85th-percentile cutoff nor the exclusion of the capital drives the results: the estimates survive an extensive set of robustness checks.[7]

This study has three main findings. First, the dzud carried a large, concentrated poverty cost that the aggregate statistics hid. In the absence of the shock, poverty in the worst-hit districts would have fallen 15.2 percentage points further during a period of exceptional national progress, and total consumption grew about 14 percent less than it would have otherwise. The headcount effect is borne almost entirely by herders, whose total effect reaches 27.6 percentage points, while the non-herder effect is smaller and statistically insignificant.

Second, the mechanism is asset destruction rather than distress sale. Herds in the worst-hit districts were about a third smaller in 2024 than at baseline; a residual accounting of the winter loss implies gross losses near half the baseline herd, partly offset by restocking. Animals died before they could be sold, overwhelming the sell-to-cope buffer that operates in milder winters, and the household's own-produced meat and dairy went with them. Decomposing the loss by income source shows that livestock income is the dominant force pushing herders below the poverty line.

[6] The HSES is normally a cross-sectional survey; the 2022/24 panel subsample was fielded specifically for these two waves and covers all 21 provinces, complementing the three-province researcher-collected panel used in prior dzud research. We exclude the capital, Ulaanbaatar, throughout.

[7] The estimates follow the expected severity gradient across alternative thresholds, hold across hazard combinations and alternative estimators, and hold with Ulaanbaatar added as an urban control. The 85th percentile isolates the genuinely severe tail of an event so widespread that official classifications placed 78 percent of the country's soums in dzud condition.

Third, the shock also reached households through food, along different routes for the two populations, and in both the burden fell on the poor. Herders faced the food shock too, and if anything more severely, since their own-produced food traveled with the herd; non-herders were spared the asset shock but not the food shock, squeezed instead through the market as lost own-production met a sharp nationwide rise in food prices.[8] Relative to less-affected districts, caloric intake fell by 16 percent across all households, and the share of non-herder households reporting any food-insecurity experience rose by roughly four-fifths. Consumption losses concentrate in the lower tail of the distribution: vulnerable herders were pushed below the poverty line, and poor non-herders fell deeper beneath it. Baseline livestock insurance is associated with smaller herder losses, though the insured sample is small and the formal test imprecise, so we read insurance as suggestively protective rather than as an established buffer. These are margins on which earlier dzud research had found non-herding households unaffected (Groppo and Kraehnert, 2016, 2017).

This paper makes three contributions to the climate-poverty literature. First, the poverty cost we estimate is, to our knowledge, the first quasi-experimental estimate of a cold extreme's effect on monetary poverty.[9] Existing quasi-experimental studies of disasters and poverty concern other hazards, including tropical storms, floods, and droughts (Baez et al., 2017; Wink Junior et al., 2024; Hill and Porter, 2017; Pape and Wollburg, 2019; Baquié and Fuje, 2020), and recent rapid assessments of mega-floods rely on ex-ante microsimulation rather than observed post-event panels (Knippenberg et al., 2024); none estimates the poverty cost of a cold-extreme, asset-destruction shock in a pastoral setting. Second, we measure the transmission channel rather than infer it: prior dzud studies identify effects from district-level shock intensity in a three-province panel, treating herder status as a subsample split (Groppo and Kraehnert, 2016, 2017), whereas our national panel estimates the herder–non-herder differential directly at the household level,

[8] Appendix Tables D8 and D6 show this margin directly: the value of own-produced consumption among poor treated non-herders fell by roughly a third, while it rose among poor non-herders in less-affected districts. With only 77 treated households in this cell, the difference is not statistically significant, and we read it as directional support rather than standalone evidence; a broader food-module measure adding food gifts falls significantly (Appendix Table D7).

[9] The closest existing evidence comes from Kyrgyzstan: Sultakeev and Petrick (2025) link recurrent harsh-winter exposure to livestock sales and consumption smoothing among livestock-owning households in a panel design, and Kimsanova et al. (2024) relate cold winters to a multidimensional poverty index through descriptive regressions. Neither estimates the effect of a realized catastrophic event on poverty measured against a national line: the design-based evidence is confined to livestock owners, and the population-wide evidence is descriptive.

with income recorded by detailed source. Third, the exposure we document reaches beyond herders, where prior dzud research treats non-herders as the unaffected comparison group whose null identifies the livestock channel (Groppo and Kraehnert, 2016, 2017). Our estimates complement this evidence from upstream: they measure the first-round income and consumption collapse that transmits the shock into the long-run outcomes prior work documents.

The remainder of the paper proceeds as follows. Section 2 sets out the pastoral economy, the dzud, and prior evidence. Section 3 describes the data and treatment, Section 4 presents the identification strategy, Section 5 reports and discusses the results, and Section 6 concludes.

2 Background

2.1 Mongolia's Pastoral Livelihoods

Livestock herding is central to rural livelihoods in Mongolia, even though mining, at 23 percent of GDP in 2022, dominates the national economy (NSO, 2022). Nearly three in ten Mongolians live in a household that keeps livestock, and outside the capital roughly a third of households herd as their primary livelihood, together managing more than 64 million head of livestock across the steppe (UNDP, 2025).[10] For these households, livestock is more than a source of cash income. It provides meat and dairy for own-consumption, stores wealth, supports access to credit, and, through dried dung, supplies a common source of heating fuel in the countryside. A severe winter can therefore compress several welfare margins at once: income, food, and assets. Herders are especially exposed because their productive wealth is held in livestock that a severe winter can kill.

The 2023/24 dzud occurred during an otherwise favorable macroeconomic period. Mining-led growth helped Mongolia reach upper-middle-income status in 2024 (World Bank, 2024), and the national poverty headcount fell from 27.1 to 15.8 percent between the survey waves.[11] This decline was driven by broad-based income growth rather than

[10]Household livestock shares are authors' calculations from the 2022 HSES, population- and household-weighted respectively. Agriculture, including livestock, contributed 13 percent of GDP in 2022, but remains the primary income source for rural households (NSO, 2022).

[11]Authors' calculations from the full HSES cross-sections, population-weighted; these reproduce the official figures (World Bank, 2025). Excluding Ulaanbaatar, the headcount fell from 32.0 to 19.8 percent; the introduction's slightly different figures, 31.5 to 20.7 percent, are computed from the panel subsample

by transfers. Real per capita income rose by close to 30 percent, with gains across the distribution, as post-pandemic labor-market recovery lifted employment and real wages, led by the public sector (World Bank, 2025). Social transfers fell in real terms over the period.[12] This macroeconomic context sharpens the question: not whether Mongolia was poor or stagnant in 2024, but why poverty progress failed to reach the households and places most exposed to the winter shock.

2.2 The Dzud Phenomenon

A dzud is a catastrophic winter on the Mongolian steppe. It occurs when winter or spring weather deteriorates to the point that livestock cannot access sufficient pasture or water, leading to widespread emaciation and mortality. Government Resolution 286 of 2015 distinguishes three main forms: a white dzud (*tsagaan zud*), in which heavy snow buries pasture; an iron dzud (*tümer zud*), in which freeze-thaw-refreeze cycles form an ice crust that prevents grazing; and a black dzud (*khar zud*), which follows summer drought and a snowless, bitterly cold winter (Government of Mongolia, 2015). These types often overlap, and the 2023/24 event combined several channels.

The National Agency for Meteorology and Environmental Monitoring (NAMEM), Mongolia's state meteorological agency, classifies each soum every ten days throughout winter, and these classifications establish the event's official scope and type.[13] Section 3.1 describes how we complement them with a measure of severity when defining treatment.

Historically, dzuds have been the main source of catastrophic loss in Mongolia's pastoral sector. The triple dzud of 1999–2002 killed 11.2 million livestock, and the 2009/10 dzud killed 10.3 million, or 23 percent of total stock (UNDP, 2025). The 2023/24 event killed 8.1 million livestock, 12.5 percent of stock, and nearly half of breeding stock miscarried or perished, making it the most severe winter in more than a decade (World Bank, 2024; UNDP, 2025).[14] The 2023/24 timeline followed the classic compound pat-

rather than the full cross-sections.

[12]The employment-to-population ratio rose from 56.5 to 57.6 percent. The Child Money Programme remained in place and Erdenes Tavan Tolgoi dividends were distributed, but several COVID-era programs ended, so total real social transfers declined.

[13]Full dzud requires all primary indicators for a given type to be met simultaneously; near-dzud requires only one primary indicator plus an auxiliary condition, such as extreme minimum temperatures (Government of Mongolia, 2015).

[14]The 2009/10 event involved larger total losses, but no household panel survey with comparable consumption modules was fielded around that dzud. The 2022 and 2024 HSES waves are the first nationally representative panel bracketing a major dzud, which is why our welfare estimates focus on

tern: summer drought in 2023 reduced hay and fodder reserves, early October–November snowfall covered pastures, and sustained cold from December through February weakened livestock that could not access forage (UNDP, 2025). Agriculture contracted by 26.7 percent in the first half of 2024 (World Bank, 2024); the household welfare cost beneath that sectoral collapse is what this paper measures.

2.3 Prior Evidence

The existing evidence on dzud impacts concentrates on non-monetary outcomes and on western Mongolia. Using a household panel from 2012–2015 covering three western provinces, this work finds that the 2009/10 dzud stunted children and reduced schooling in herding households, with effects persisting for three to four years (Groppo and Kraehnert, 2016, 2017). Insurance purchasers recovered their herds faster (Bertram-Hümmer and Kraehnert, 2018), and affected herders experienced persistent asset losses, exits from pastoralism, and permanent out-migration consistent with poverty-trap dynamics (Lehmann-Uschner and Kraehnert, 2018; Roeckert and Kraehnert, 2022). Livestock wealth inequality also rose sharply and remained elevated four years later (Kakinuma et al., 2024). These findings align with the broader natural-disaster literature, where households often smooth shocks by cutting non-food spending, especially health and education, at a long-run human-capital cost (Karim and Noy, 2016). Across these studies, impacts concentrate in herding households; non-herders are the unaffected comparison group, and that herder–non-herder contrast is the identifying variation of prior dzud research.

This paper adds the monetary layer that precedes those scars. The stunting, interrupted schooling, and slow asset recovery documented in that literature must first pass through a collapse in income and consumption, and this first-stage margin is what our panel measures: poverty, income by detailed source, and consumption before and after the event, with national coverage that includes the eastern provinces where the 2023/24 dzud struck hardest. We can therefore estimate the poverty cost directly and separate livestock-income losses from other income sources, filling in the opening link of a causal chain whose later stages that literature has traced. Measuring welfare in money also changes who counts as affected. Where prior outcomes locate the damage within

2023/24.

herding households, the margins we observe extend beyond the herd: non-herders are hit through consumption and food security, including a food-access squeeze among poorer households, a form of exposure that non-monetary outcomes could not pick up.

3 Data and Measurement

3.1 Treatment Definition

NAMEM's official classifications record the type of dzud a soum experienced, not its severity. The typology introduced in Section 2.2 registers whether the conditions of a given dzud form were present, and the 2023/24 winter brought such conditions to most of the country: NAMEM classified 257 of 330 non-Ulaanbaatar soums, or 78 percent, as experiencing actual dzud on at least one date, and the lower near-dzud category raises coverage to 91 percent (Appendix Figure B1; authors' calculations from NAMEM ten-day dzud classifications, winter 2023/24). Our treatment therefore measures the dimension the typology does not record, severity, drawing on the continuous weather data underlying those conditions to compare the most severely affected soums with soums that experienced more moderate winter conditions.

We construct treatment at the soum level from two weather channels over the November–February winter window: temperature and snow cover. The soum is the relevant spatial unit for three reasons. It is the level at which NAMEM classifies the dzud, the level used in prior dzud exposure measures (Groppo and Kraehnert, 2016, 2017), and the administrative unit around which pastoral land use is organized. Herder families hold long-term winter-campsite contracts, while soum governments regulate seasonal movements under Mongolia's Land Law (Batdelger et al., 2025). Long-distance *otor* movements of 10–200 km occur as a response to forage scarcity, but they cannot reclassify a household: treatment is fixed at the 2022 baseline soum, so neither seasonal grazing movement nor post-dzud relocation changes a household's treatment status.[15]

For each soum s, the temperature deviation is the difference between the 1992/93–2022/23 historical winter mean and the 2023/24 winter mean:

$$\Delta T_s = \bar{T}_s^{\text{base}} - \bar{T}_s^{2023/24} \tag{1}$$

[15]We confirm robustness to migration in the long-term-stayer subsample (Appendix F.2).

The sign is chosen so that positive values indicate a colder-than-normal winter. Both terms come from ERA5 reanalysis aggregated to the soum.[16] The snow-cover anomaly is constructed analogously:

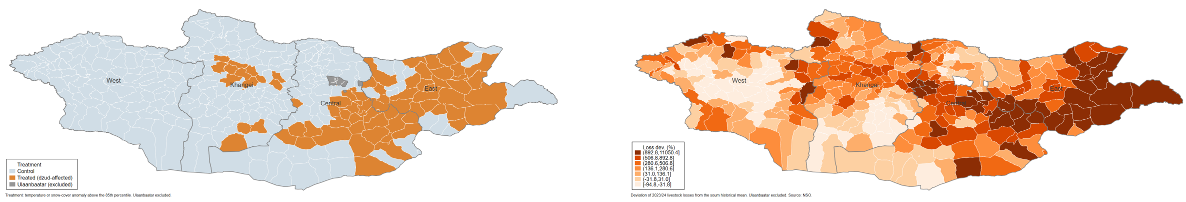
$$\Delta S_s = \bar{S}_s^{2023/24} - \bar{S}_s^{\text{base}} \tag{2}$$

Positive values denote more snow cover than normal. We rank both deviations across non-Ulaanbaatar soums in 2023/24, mapped in Appendix Figure B2, and define a soum as treated if either deviation is in the top 15 percent:

$$\text{Treated}_s = \mathbf{1}\left[\Delta T_s \geq \Delta T^{P85} \ \text{ or } \ \Delta S_s \geq \Delta S^{P85}\right]. \tag{3}$$

This union captures both the cold and snow-burial channels in a single binary measure. It classifies 79 of 330 non-Ulaanbaatar soums, or 23.9 percent, as treated (Appendix Table B1, Panel B). The treatment produces balanced groups on most baseline characteristics (Appendix Table B2), stable pre-trends (Figure C1), and results that are robust to temperature-only, snow-only, drought-interaction, and alternative percentile definitions (Appendix F.5).

Figure 1 shows the preferred treatment definition alongside the livestock losses it predicts.



(a) Treatment: Temperature or Snow P85

(b) Livestock Loss Deviation

Figure 1: Treatment Definition and Livestock-Loss Validation

Notes: Panel (a) shows the preferred binary treatment, temperature or snow P85 (79 treated, 251 control). Panel (b) shows the percentage deviation of 2023/24 livestock losses from each soum's 2015–2023 historical mean, using NSO administrative data. Both maps exclude Ulaanbaatar. Appendix Figure B1 maps two further benchmarks: the official NAMEM dzud classification and herders' self-reported dzud shocks.

The validation evidence supports the severity definition. Livestock losses (Panel b) concentrate in the eastern provinces of Dornod, Sukhbaatar, Khentii, and Dornogovi,

[16] Daily ERA5 temperature is overlaid on an H3 hexagonal mesh and averaged within each soum boundary. Snow cover comes from the MODIS satellite product: snow-cover fraction, the share of land under snow rather than snow volume. Its baseline is the 11 winters 2012/13–2022/23, shorter than the temperature baseline because MODIS begins in 2012.

where the weather-based treatment also falls (Panel a), and herders' self-reported dzud shocks cluster in the same soums (Appendix Figure B1).

NAMEM's official classification is much broader (Appendix Figure B1), and the breadth is by design: a hazard-typology instrument rather than a severity ranking, it flags whether the mechanism of a given dzud type, such as white or iron dzud, is present, with thresholds set for early warning and therefore deliberately sensitive. Our treatment instead isolates the upper tail of the severity distribution, not the mere presence of dzud conditions.[17]

3.2 Household Data and Outcomes

We use the Household Socio-Economic Survey (HSES), conducted biennially by the National Statistics Office (NSO). The HSES uses monthly fieldwork and stratified two-stage cluster sampling; the 2022 wave covers about 23,000 households, 18,318 of them outside the capital. The 2022 and 2024 rounds include a representative panel subsample of 2,128 households, or 4,256 household-year observations, across 175 soums and all 21 provinces. The 2022 wave measures households before the dzud; the 2024 wave measures them after it. All estimation uses population weights and clusters standard errors at the baseline soum level.

We exclude Ulaanbaatar because the paper studies a livestock-based winter shock in provincial Mongolia. Ulaanbaatar has no meaningful pastoral sector, so the main transmission channel does not apply. Three of its districts also cross the temperature threshold, which creates a mechanical problem: including the capital would place nearly half of the national population into three urban "treated" districts and swamp the rural effect.[18] The resulting sample covers 330 non-Ulaanbaatar soums, which we refer to as *provincial Mongolia*.[19]

Appendix Table B1 summarizes sample construction and treatment assignment.

[17]NAMEM's near-universal coverage in 2023/24 means most of the country experienced dzud-type conditions, not that conditions were uniformly severe. The continuous measures rank severity within that event; Table 1 quantifies the treated–control gap in livestock losses.

[18]Bayanzurkh, Songinokhairkhan, and Bayangol are the only districts of the capital's nine that cross the temperature threshold. Their inclusion would not alter the treatment landscape outside the capital; it would, however, concentrate the bulk of the treated population in these urban districts (authors' calculations from ERA5 soum-level treatment data and HSES population weights).

[19]The urban/rural classification is at the soum level: of the 175 panel soums, the 21 aimag centers are urban and the remaining 154 are rural. Rural soums house about 60 percent of the sample population after population weighting (authors' calculations from HSES population weights).

Of the 2,128 panel households, 530 reside in treated soums. The panel treatment share, 24.9 percent, closely matches the full non-Ulaanbaatar soum share, 23.9 percent. Attrition is uncorrelated with treatment (Appendix B.1).

The primary outcome is **poverty status**, an indicator for whether household consumption falls below the national poverty line of 16,882 MNT per adult equivalent (PAE) per day in 2024 prices, approximately \$4.89.[20] This is the headcount measure, FGT_0, from the Foster–Greer–Thorbecke (FGT) poverty family (Foster et al., 1984). For scale, Mongolia's national poverty rate was 27.1 percent in 2022 (World Bank, 2025), so the effects we estimate are large relative to the national stock of poverty.

Supplementary outcomes carry the incidence and mechanism analysis. The poverty gap, FGT_1, and poverty severity, FGT_2, enter the distributional analysis alongside quantile treatment effects (Section 5.2; definitions in Appendix Table A1 and Appendix E.1); these measures take the paper beyond the headcount to the depth and incidence of losses. Log total consumption PAE, the welfare aggregate against which poverty is assessed, is the outcome for the quantile analysis. We use log income PAE to study mechanisms, and log food consumption PAE, caloric intake, and an indicator for reporting any of five experiences from the FAO Food Insecurity Experience Scale (FIES) to study the food channel.

4 Identification Strategy

The design is a two-period, binary-treatment difference-in-differences setting. Households are observed before and after a single winter shock, and treated soums are compared with less-exposed soums. This structure avoids the negative-weighting and forbidden-comparison problems that can arise in two-way fixed effects designs with staggered treatment timing and heterogeneous effects (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021).[21] Following the practical guidance of Baker et al. (2026), we define the target parameter, state the identifying assumptions, select the estimator, and evaluate the design.

[20] Mongolia's poverty line is an absolute, consumption-based measure set by the NSO and adjusted for regional and seasonal price differences; the 2022 threshold is updated to 2024 prices. PAE uses the modified OECD equivalence scale.

[21] Those pathologies arise under staggered adoption, where already-treated units can be used as controls for later-treated units. They do not arise in a single-timing 2×2 design.

4.1 Design

Target parameter. The target is the average treatment effect on the treated (ATT). With $Y_{is}(1)$ and $Y_{is}(0)$ denoting post-period potential outcomes under severe dzud exposure and less-severe exposure for household i in soum s,

$$\text{ATT} \equiv \mathbb{E}_w[Y_{is}(1) - Y_{is}(0) \mid \text{Treated}_s = 1] \,, \tag{4}$$

where $\mathbb{E}_w$ is the population-weighted expectation.[22] Because the control group also experienced some winter stress, the ATT should be read as the incremental effect of severe exposure relative to moderate exposure, not as the effect relative to a no-dzud world. For the triple difference, the target is $\text{ATT}_{\text{TD}} \equiv \text{ATT}_{\text{herder}} - \text{ATT}_{\text{non-herder}}$, the additional welfare loss borne by herders relative to their non-herder neighbors in the same soums. Caron (2025) calls this parameter the differential ATT (DATT).

Specification. We estimate linear models throughout. For the binary poverty outcome, this is a linear probability model; for income, consumption, and FGT gap measures, it is a linear outcome model. Our preferred specification uses household and year fixed effects, with baseline covariates, region, and interview quarter interacted with Post_t:

$$Y_{ist} = \alpha_i + \lambda_t + \beta\,(\text{Treated}_s \times \text{Post}_t) + (X_{is,0} \times \text{Post}_t)'\delta + \varepsilon_{ist}, \tag{5}$$

where α_i and λ_t are household and year fixed effects and $X_{is,0}$ collects fourteen pre-treatment characteristics measured in 2022.[23] The covariate-by-post terms allow the 2022–2024 trend to differ by baseline characteristics, so identification is based on conditional parallel trends. Year fixed effects absorb shocks common to all households, including mining-led growth and broad wage gains. Region-by-post interactions absorb differential regional trends, which matters because the 2023/24 dzud was geographically

[22] Population weighting aligns the ATT with how poverty is reported, as the share of the population below the line. We report household-weighted and unweighted estimates as robustness checks (Appendix F.2).

[23] Head age, gender, and education; household size; urban status; herder status; inverse hyperbolic sine (IHS) of asset resale value and baseline livestock holdings in *bod* units, the Mongolian large-livestock equivalent; insurance status; number of working members; and baseline income shares from wage, agricultural, and business income. Software: TWFE via `reghdfe` (Correia, 2016), DR-DD via `drdid` (Sant'Anna and Zhao, 2020), sensitivity via `HonestDiD` (Rambachan and Roth, 2023), and coefficient stability via `psacalc` (Oster, 2019).

concentrated in the East. Quarter-by-post interactions absorb the HSES rolling-interview calendar, since households interviewed early in 2024 may report a partly pre-dzud reference window. Standard errors are clustered at the baseline soum level, the level at which treatment is assigned, giving 175 clusters in the panel.

We report DD estimates without and with covariates in Columns 1–2 of Table 2, and the corresponding triple-difference estimates in Columns 3–4. In a balanced two-period panel, the no-covariate TWFE estimate equals the unconditional OLS level-difference form. Reporting both specifications also addresses a concern with covariate-adjusted TWFE: imposing common covariate slopes can distort the ATT when treatment effects vary with covariates (Baker et al., 2026; Sant'Anna and Zhao, 2020). In practice, the estimates are stable with and without covariates, baseline covariates are balanced, and inverse-probability-weighted and doubly robust DD estimators give similar results (Section 5.5).

The remaining concern is time-varying local shocks correlated with severity, such as differential government response, road disruption, market disruption, or migration. The triple difference addresses this concern by adding baseline herder status and the relevant lower-order interactions:

$$\begin{aligned} Y_{ist} = \alpha_i + \lambda_t + \beta_1 \left(\text{Treated}_s \times \text{Post}_t\right) + \beta_2 \left(\text{Post}_t \times \text{Herder}_{i,0}\right) \\ + \gamma \left(\text{Treated}_s \times \text{Post}_t \times \text{Herder}_{i,0}\right) + \varepsilon_{ist}. \end{aligned} \tag{6}$$

Here $\text{Herder}_{i,0}$ is herder status in 2022. The coefficient β_1 is the effect for non-herders, γ is the additional effect for herders, and $\beta_1 + \gamma$ is the implied total effect for herders. Under the parallel-gap-trends assumption stated below, γ estimates the DATT (Olden and Møen, 2022).[24] The TD differences out any soum-by-year shock common to herders and non-herders in the same place; the remaining threat is therefore a factor that differentially affects herders in treated soums relative to herders in control soums, beyond the livestock channel itself.

Herder status is fixed at baseline to avoid post-treatment classification. The dzud can itself push households out of herding, so current occupation is potentially endogenous. Between waves, 244 panel households (11.5 percent of the sample) changed herder status:

[24] We target the raw DATT, not Caron's covariate-adjusted CDATT. Conditioning on the characteristics that distinguish herders, especially livestock dependence, would purge the channel through which the dzud operates.

138 exited herding and 106 entered (authors' calculations from the HSES 2022/2024 panel employment module). Because the switching window straddles the dzud, all subgroup analyses use 2022 herder status.

4.2 Identifying Assumptions and Evidence

The strategy rests on four assumptions: parallel trends, parallel gap trends for the triple difference, no anticipation, and SUTVA. We state each assumption in turn, followed immediately by the evidence bearing on it. The subsection closes with direct validation that the treatment definition captured the livestock mortality that defines a dzud.

Parallel trends. Absent severe dzud exposure, treated and control soums would have experienced the same average change in outcomes:

$$\mathbb{E}_w[\Delta Y_{is}(0) \mid \text{Treated}_s = 1] = \mathbb{E}_w[\Delta Y_{is}(0) \mid \text{Treated}_s = 0]\,. \tag{7}$$

The content of this condition depends on the outcome's functional form. For the binary poverty outcome, parallel trends mean that poverty rates in treated and control soums would have moved by the same number of percentage points; for log income and consumption, they would have grown at the same percentage rate. Each estimate is therefore judged against its own counterfactual metric: changes in percentage points for poverty, growth rates for income and consumption.

A priori, the assumption is plausible because treatment is assigned by weather. Exogeneity here means that weather realizations are random draws over time from a soum's own climate distribution (Dell et al., 2014, pp. 744–745): no household action or characteristic causes a soum's winter to cross the 85th percentile of its own history. The randomness, however, is over time, not across space: which soums are exposed is not random, because severe winters strike particular kinds of places, and places differ in composition, region, and livelihoods. Exposed and unexposed soums could therefore trend differently for reasons unrelated to the dzud. This is why the assumption requires evidence of its own.

The assumption is a statement about counterfactual trends and is not directly testable (Baker et al., 2026): it concerns the path treated soums would have followed after the dzud, not the path they followed before it, so no pre-period test can verify it. Pre-dzud stability is nonetheless supportive evidence, since a treated group that tracked its controls before the shock is more plausibly one that would have continued to do so. The panel itself contains no pre-period, consisting of the 2022 and 2024 waves alone, so we test for differential pre-dzud trends using our second-best source, HSES pooled cross-sections from 2020 and 2022 restricted to the 175 panel soums (Appendix C).[25] For poverty, treated and control soums decline in parallel over 2020–2022 (Appendix Figure C1). In the figure, identification requires the two groups' poverty rates to move together before the shock, and they do; the treatment effect itself is estimated within panel households (Figure 2 and Table 2). As Roth (2022) cautions, failing to reject a pre-trend does not prove parallel trends, so Section 5.5 complements the visual evidence with formal sensitivity analysis following Rambachan and Roth (2023).

Covariate balance provides a second check on comparability. Appendix Table B2 compares 2022 characteristics across treatment groups in three panels: baseline outcomes, predetermined covariates, and livelihood characteristics. Following Baker et al. (2026), we restrict the balance and propensity-score covariates to predetermined characteristics, matching the inverse-probability-weighting (IPW) and doubly robust (DR-DD) specifications. Predetermined characteristics are well balanced. Age, education, household size, assets, livestock, and livelihood composition all have Imbens–Rubin normalized differences below 0.25 (Imbens and Rubin, 2015). Baseline food consumption and insurance take-up are modestly higher in treated areas. Treated and control households are thus structurally similar: herder shares are nearly identical (39 and 40 percent), and the differences that remain are in welfare levels rather than composition.

Levels are absorbed by the household fixed effects, differential trends are the subject of the pre-trend evidence above and the placebo tests below, and the reweighting estimators in Appendix Table F2, IPW and doubly robust DD on fifteen baseline covariates, address the same comparability concern that motivates matching.

The main imbalance is geographic (Appendix Table B2). Half of treated households reside in the Eastern region, compared with 5 percent of controls; no treated soums

[25] The pre-trend evidence therefore comes from a different sample than the treatment effects, though the same soums. Households differ across waves, so the check speaks to whether treated and control *soums* were on parallel paths before the dzud, not whether the panel households themselves were. Shifts in within-soum composition could obscure or mimic a differential trend, so we read this evidence as suggestive.

lie in the Western region. This pattern reflects the climatology of the 2023/24 dzud, not sampling error. We address it in three ways: region-by-post interactions absorb region-specific trends, the Western-aimag drop confirms robustness (Section 5.5), and the TD compares herders and non-herders within the same soums. Interview timing is balanced across treatment groups in 2022 (Appendix Table B3, Panel A). In 2024, Q1 shows a borderline imbalance ($d = -0.26$; Panel B), with fewer treated households interviewed in January–March before the dzud's impact had fully materialized. Quarter-by-post interactions absorb this timing difference, and the Q1-exclusion check confirms robustness (Appendix Table F1).

Parallel gap trends for the TD. The triple difference requires the herder–non-herder outcome gap to trend similarly across treated and control soums in the absence of the shock (Olden and Møen, 2022). Writing H for herders and NH for non-herders, the condition is:

$$\begin{aligned} &\underbrace{\Delta\big[E(Y_0 \mid \text{Treated}, H) - E(Y_0 \mid \text{Treated}, NH)\big]}_{\text{change in treated-soum herder gap}} \\ &= \underbrace{\Delta\big[E(Y_0 \mid \text{Control}, H) - E(Y_0 \mid \text{Control}, NH)\big]}_{\text{change in control-soum herder gap}}. \end{aligned} \tag{8}$$

This condition is weaker than requiring parallel trends separately for herders and non-herders. Herder outcomes may be more volatile than non-herder outcomes, and treated soums may differ from control soums, as long as the herder–non-herder gap would have evolved similarly across treatment groups. Intuitively, the non-herder DD removes bias from the herder DD if that bias is similar for both groups (Olden and Møen, 2022). Appendix Figure C2 provides the corresponding evidence: the herder–non-herder poverty gap differential between treated and control soums is stable over 2020–2022, consistent with this condition.

No anticipation. The 2022 outcomes must not be affected by the 2023/24 dzud. This is plausible on timing grounds: the shock arrived with the winter of 2023/24, more than a year after the 2022 fieldwork, and herders cannot anticipate the severity of a specific winter two seasons in advance. The public response that followed the realized shock is a post-period matter, discussed under SUTVA below.

SUTVA: control status and spillovers. The stable unit treatment value assumption requires that treatment is well defined for every soum and that one soum's exposure does not affect another soum's outcomes. On the first, control soums experienced less-severe winters, not the absence of dzud-type weather (Section 3.1). The DD therefore estimates the effect of extreme conditions relative to moderate conditions. If control outcomes were also depressed by winter stress, the estimates are attenuated relative to a true no-dzud benchmark.[26]

On the second, spillovers appear more likely to attenuate than inflate the results. Treated-area scarcity could, in principle, raise control-herder revenue, but the price evidence points the other way: live-animal selling prices fell during the dzud as distress selling depressed them (UNDP, 2025), and per-unit prices among herders who sold in both years were flat to slightly higher (Section 5.3). Partial treatment of controls also pushes against finding large effects.

The public response to the realized shock works in the same direction. NAMEM issued formal early warnings in December 2023, and the government and humanitarian actors mounted a large anticipatory and early-recovery response over winter 2023/24 and spring 2024, totaling about 51.1 billion MNT (about $15 million) of state aid across twelve categories, including hay, fodder, food, road clearing, and *otor* support, alongside Red Cross, UN-cluster, and bilateral assistance (UNDP, 2025). These responses may have attenuated the shock, but they did not offset it: more than eight million livestock died. We therefore interpret the DD as a conservative estimate of severe exposure relative to a no-dzud counterfactual, net of early-warning-triggered assistance.

Treatment validation: did the dzud operate as expected? The defining feature of a dzud is livestock mortality. Table 1 links the continuous climate variables to soum-level livestock losses, measured as the standardized deviation of 2024 *bod* losses from each soum's 2015–2023 historical mean. Treated soums averaged 8.7 standard deviations above historical-mean losses, compared with 1.7 standard deviations for controls (authors'

[26] Control herders also destocked under pressure, selling about 40 percent more livestock while their herd fell about 10 percent (authors' calculations from the HSES livestock module, control-herder subsample). This coping is itself evidence of milder harm and reinforces the lower-bound reading: distress selling converts future assets into current consumption, so control outcomes sit below their no-dzud counterfactual. The only offsetting channel, controls gaining through depressed livestock prices or cheap restocking, is inconsistent with the near-zero annual price difference-in-differences and with controls' own net herd decline. In the worst-hit areas, the herd decline was much larger and far exceeded what sales and own-consumption can account for (Section 5.3).

calculations from the NSO soum-level livestock-loss data; replication package). A one-degree-Celsius colder winter raises the loss z-score by 3.0 standard deviations, and a one-percentage-point increase in snow cover raises it by 0.10 standard deviations. Each channel predicts mortality on its own (Columns 1 and 2), and both survive when entered jointly (Column 3). With region fixed effects (Column 4), the temperature coefficient falls as expected because cold exposure was regionally concentrated, while snow cover remains significant; the climate treatment thus predicts the livestock mortality that defines a dzud.

Table 1: Climate Conditions and Livestock Loss

	Livestock Loss (z-score)			
	(1)	(2)	(3)	(4)
Temperature deviation (°C)	4.453***		2.986***	2.135
	(0.884)		(0.753)	(1.306)
Snow cover anomaly (pp)		0.168***	0.097**	0.096**
		(0.041)	(0.040)	(0.042)
Region FE	No	No	No	Yes
R^2	0.123	0.111	0.146	0.203
Observations	330	330	330	330

Notes: The dependent variable is the soum-level livestock loss z-score: the standardized deviation of 2024 *bod* losses from the 2015–2023 historical mean, computed from NSO administrative data. Temperature deviation is measured in degrees Celsius, with positive values indicating colder-than-normal conditions (Eq. 1); snow-cover anomaly is measured in percentage points, with positive values indicating more snow cover (Eq. 2). Treatment is defined as a winter temperature deviation or snow-cover anomaly above the 85th percentile of the historical soum distribution (Eq. 3). Region fixed effects cover four regions: West, Highlands, Central, and East. The sample excludes Ulaanbaatar and is consistent across columns. Robust standard errors are in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

5 Results and Discussion

5.1 The Poverty Cost

The dzud froze poverty reduction where it struck while poverty fell steeply everywhere else, and Figure 2 shows this in raw means. Between 2022 and 2024, mining-led growth cut the national poverty headcount from 27.1 to 15.8 percent; in our estimation sample, provincial Mongolia outside the capital, it fell from 31.5 to 20.7 percent. Control soums shared fully in that decline, while poverty in treated soums barely moved. The dashed line applies the control-group decline to the treated baseline; the distance between this counterfactual and the observed treated rate is the poverty cost of the shock.

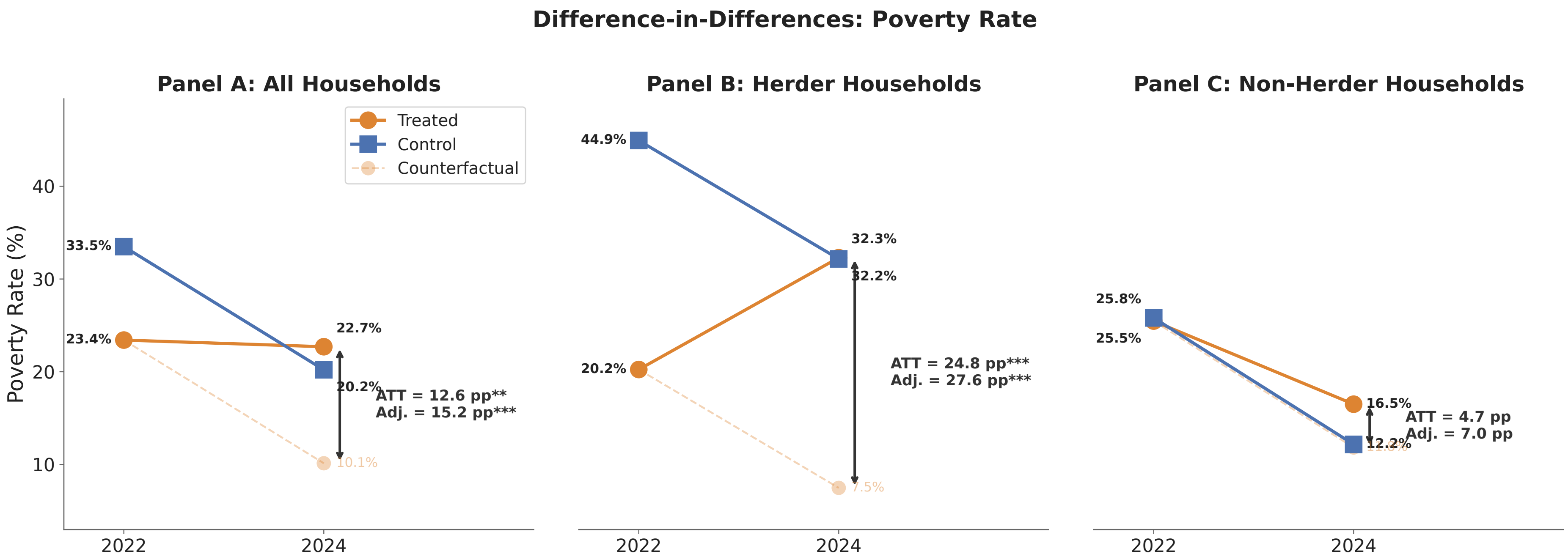


Figure 2: Poverty Rates by Treatment Group (2022–2024)

Notes: Population-weighted poverty rates. The dashed line shows the treated-group counterfactual (treated baseline plus the control change). In each panel, *ATT* is the raw means-based difference-in-differences and *Adj.* the covariate-adjusted two-way fixed-effects estimate from Table 2, both in percentage points. The vertical axis is shared across panels and shown on Panel A. Panels B and C decompose by baseline (2022) herder status. Balanced panel, provincial Mongolia. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

The subgroup panels locate the cost. In treated soums, poverty rose among herders while it continued falling among non-herders; in control soums it fell for both groups (Panels B and C). What reads as aggregate stagnation in treated areas is therefore a sharp reversal for livestock-dependent households, offset in the average by their neighbors' continued progress.

Table 2 quantifies the gap. In the preferred covariate-adjusted specification (column 2), poverty in treated soums would have fallen 15.2 percentage points further in the absence of the dzud. The estimate should be read as forgone poverty reduction rather than an absolute rise: poverty in treated soums edged down from 23.4 to 22.7 percent over these two years, while in control soums it fell from 33.5 to 20.2 percent. The shock did not so much impoverish treated soums as hold them in place while the rest of provincial Mongolia moved.

Table 2: Effects of Dzud Exposure on Poverty

	Difference-in-differences		Triple difference	
	(1)	(2)	(3)	(4)
Treated × Post	0.126**	0.152***	0.047	0.070
	(0.049)	(0.054)	(0.044)	(0.045)
Treated × Post × Herder			0.201**	0.206**
			(0.087)	(0.084)
Controls × Post	No	Yes	No	Yes
Household + Year FE	Yes	Yes	Yes	Yes
Observations	4,256	4,254	4,256	4,254
Control mean, 2022	0.335	0.335	0.335	0.335
Herders			0.449	0.449
Non-herders			0.258	0.258

Notes: The dependent variable is a binary poverty indicator. In the triple-difference columns, Treated × Post is the effect for non-herders and Treated × Post × Herder the additional effect for herders; their sum, 0.276 (SE 0.089, $p < 0.01$) in column 4, is the total herder effect. All columns include household and year fixed effects; columns 2 and 4 add 2022 baseline covariates interacted with post (head age, gender, and education; household size; urban status; herder status; IHS asset resale value; IHS livestock holdings in bod units; insurance status; number of working members; and wage, agricultural, and business income shares), plus region × post and quarter × post. The two-observation drop is one household with undefined income shares. The lower panel reports the population-weighted 2022 control-group poverty rate, overall and by baseline herder status. Standard errors, clustered at the baseline soum level, in parentheses; population weights. Full main effects and OLS level-difference specifications are available in the replication package. * p<0.10, ** p<0.05, *** p<0.01.

These magnitudes are large against the relevant baselines. The area-level effect equals nearly half of the 33.5 percent control-group poverty rate in 2022, and the total herder effect of 27.6 percentage points (Table 2, notes) exceeds half of the 44.9 percent baseline rate among control herders; the insignificant non-herder estimate is measured against a lower 25.8 percent base. Figure 2 translates the herder coefficient into levels: treated-herder poverty climbed from about 20 to 32 percent while control-herder poverty fell from about 45 to 32 percent. The dzud did not merely slow herders' exit from poverty; it reversed it.

5.2 Who Bears It

The area-level estimate conceals who bears the cost. Columns 3–4 of Table 2 assign the effect to livelihoods. The coefficient on Treated × Post, the effect for non-herders, is 7.0 percentage points in column 4 and statistically insignificant. The herder interaction adds a further 20.6 percentage points, for a total herder effect of 27.6 percentage points, significant at the 1 percent level (Table 2, notes). The herder effect is thus roughly four times the non-herder effect, and only the herder effect is statistically distinguishable from zero.

The muted non-herder headcount does not mean non-herders escaped. About three-quarters of control non-herders already stood above the poverty line in 2022 (Table 2, lower panel), and national growth carried most of them further from it, so a moderate welfare loss rarely produced a poverty spell. The remainder of this section shows where the non-herder loss surfaces instead, lower in the consumption distribution; Section 5.3 traces it to food rather than measured income. The headcount effect is herder-driven; the welfare loss is broader.

The headcount understates the dzud's welfare cost because it registers only crossings of the poverty line. Total consumption grew about 14 percent less in treated than in control areas, with an additional herder-specific loss of about 13 percent (Appendix Table D1), and losses of that size must fall on many households the headcount never counts, some already below the line and some comfortably above it. The question is where in the welfare distribution the losses landed.

FGT poverty measures, which register whether households are poor (headcount), how far below the line they fall (gap), and how unequally (severity), give a first answer: the two groups contributed through different margins (Appendix Table E1). The herder differential is concentrated in the headcount, households pushed across the line. The non-herder effect appears instead in depth: the poverty gap and severity rise significantly for non-herders even though their headcount response is small and insignificant, so non-herder losses deepened existing poverty more than they created new poverty. The pattern matches the asymmetry built into simulation-based disaster assessments, in which well-being losses concentrate among the poor by the structure of the assumed welfare function (Hallegatte et al., 2017; Walsh and Hallegatte, 2020); here the concentration is estimated from realized post-shock outcomes.

Unconditional quantile treatment effects sharpen the picture (Table 3). The quantiles are fixed points of the national consumption distribution, from Q15 among the deep poor to Q90 near the top, so a given quantile marks the same welfare rung in every panel. Herder losses are large and significant through the lower half: 51 log points at Q15, 35 at Q30, and 24 at Q50 (the last at the 10 percent level), with nothing detectable at Q70 or Q90. The effect near Q30, the quantile closest to the poverty line, is the distributional counterpart of the headcount result; the larger effect at Q15 shows that already-poor herders fell further still.[27]

Non-herder losses are smaller and sit low in the distribution. They are significant at Q15 (-0.189) and Q50 (-0.144) but nowhere large enough to generate much headcount response, consistent with the FGT evidence that the dzud deepened poverty among poorer non-herders without pushing many across the line. Higher up the distribution, the point estimates are negative but indistinguishable from zero.

Food, not income, is the source of this lower-tail non-herder loss. The non-herder income decomposition shows no robust income decline among the poor (Appendix Table D6), whereas food-consumption quantiles show significant non-herder losses through most of the distribution: largest at Q15 and Q30 (about 30 log points each), present at Q50, significant again at Q90, and absent only at Q70 (Appendix Table E2). The largest proportional losses sit at the bottom, exactly where the FGT results locate the deepening of poverty. The non-herder margin is not a muted copy of the herder livestock shock; it is a food-access shock that weighs most heavily on poorer households (Appendix Table D7). Taken together, the incidence splits by margin: the dzud pushed herders across

[27]These are forgone-growth effects, not absolute declines: consumption grew broadly over 2022–2024, so the -0.351 at Q30 means the rung near the poverty line would have grown about 35 log points more absent the dzud.

the poverty line, an extensive-margin effect, while it deepened the poverty of already-poor non-herders, an intensive-margin effect.

Table 3: Unconditional Quantile Treatment Effects on Log Consumption

	(1) Q15	(2) Q30	(3) Q50	(4) Q70	(5) Q90
Panel A: DD – All Households					
Treated × Post	-0.316***	-0.174**	-0.161**	-0.045	-0.057
	(0.085)	(0.071)	(0.076)	(0.074)	(0.067)
Observations	4,254	4,254	4,254	4,254	4,254
Panel B: DD – Non-Herders					
Treated × Post	-0.189***	-0.082	-0.144**	-0.087	-0.110
	(0.048)	(0.060)	(0.056)	(0.074)	(0.102)
Observations	2,992	2,992	2,992	2,992	2,992
Panel C: DD – Herders					
Treated × Post	-0.513***	-0.351***	-0.243*	-0.037	0.007
	(0.177)	(0.135)	(0.133)	(0.120)	(0.099)
Observations	1,262	1,262	1,262	1,262	1,262
Control mean (2022, MNT/day)			22,846		

Notes: Unconditional quantile treatment effects from recentered influence function (RIF) regressions (Firpo et al., 2009) on the national consumption distribution (Appendix E.2), using the covariate-adjusted specification of Table 2. Coefficients are in log points and measure forgone consumption growth relative to the counterfactual, not absolute declines. Panels B and C decompose the aggregate shift in Panel A by baseline herder status. "Control mean (2022, MNT/day)" is the population-weighted control-group mean of daily consumption PAE. Standard errors, clustered at the baseline soum level, in parentheses; population weights. * p<0.10, ** p<0.05, *** p<0.01.

5.3 The Two Channels

Two channels lie behind this incidence pattern, livestock and food, and we take livestock first. Poverty effects concentrated on herders point to livestock, but pointing is not measuring. Prior dzud studies infer the livestock channel from the concentration of schooling, child-growth, and asset effects among herding households (Groppo and Kraehnert, 2016, 2017); the panel's income module lets us weigh the channel directly. The question is how much of the herder welfare loss ran through livestock income itself.

The answer is most of it. Relative to control herders, treated-herder total income fell by about 9.4 million MNT per household per year, roughly $2,700, a gap significant at the 1 percent level (annual, 2024 prices; Appendix Table D3). Livestock income, which supplied two-thirds of herder income at baseline (66.7 percent; Appendix Table D3, memo), accounts for 71 to 83 percent of this loss, depending on whether the decomposition adjusts for baseline covariates. The loss arrived through the market and the kitchen alike: sales of livestock and byproducts make up about 77 percent of the livestock-income decline, and forgone own-consumption the remaining 23 percent (Appendix Table D4). Wage income also grew about 2.0 million MNT less for treated herders, significant at the 10 percent level, but no reading of the decomposition displaces livestock as the dominant mechanism.[28]

One caveat qualifies the shares. Income is the outcome least robust to parallel-trends violations (Appendix C), so we treat the decomposition as indicative of composition rather than exact. The mechanism does not hinge on it: the herd-mortality and stock evidence points to the same channel through physical rather than income measures (Table 1; Appendix Table D5).

What collapsed was the asset itself. Treated herders' herds shrank by 33 percent on net between 2022 and 2024, against 10 percent among control herders (Appendix Table D5). The net change understates the winter toll: a residual accounting of the loss, the baseline herd plus sales and own-consumption less the surviving herd, implies gross winter losses near half the baseline herd, partly offset by restocking. On that accounting, most animals that left treated herds died before they could be sold. The share of treated herders selling any livestock also fell, by 10.5 percentage points against 0.7 among controls, a decline that is imprecise but suggestive of some households exiting livestock markets entirely.

A sustained price shock, the other candidate mechanism, finds little support in the annual data. Among herders who sold in both years, decomposing sales revenue into price and quantity attributes the change to quantity, although both margins are imprecisely estimated (Appendix Table D4, Panel B). Administrative price series do record a distress-pricing episode, but a short-lived one confined to the peak winter months, which the annual survey recall, dominated by autumn sales after prices recovered, largely averages out.[29]

[28] A natural concern is that relief transfers surged into treated areas and masked part of the loss. They did not: the difference-in-differences on non-labor income (pensions, public transfers, and capital income) is insignificant for herders and non-herders alike (Appendix Table D3), and aggregate public transfers *fell* over the panel, from a 16.9 to an 11.3 percent income share, as COVID-era programs expired (authors' calculations from the HSES 2022/2024 panel income module). Reactive transfers would, if anything, bias the income and consumption DDs toward zero, making them lower bounds.

[29] NSO administrative wether-sheep prices confirm that intensive-margin distress pricing occurred, but only in a narrow window: the treated–control price difference-in-differences is about −7 percent over February–April 2024 and roughly zero over the full year, consistent with UNDP (2025)'s account

The dominance of the quantity margin clarifies what livestock can and cannot buffer. In moderate shocks herders sell livestock to smooth consumption, as Kyrgyz livestock farmers do in harsh winters (Sultakeev and Petrick, 2025), and less-affected Mongolian herders did some of this in 2023/24.[30] The winter overwhelmed that margin in the treated soums: herds fell far faster than sales rose, and the residual accounting attributes most of the decline to winter deaths rather than planned destocking. The poverty effect therefore reflects the destruction of a productive asset, not the orderly sale of one to protect consumption.

Food is the shock's second channel, and it reached both populations. For herders, the caloric loss traveled with the herd and belongs to the livestock channel itself; for non-herders, whose measured income loss is small and statistically insignificant (Appendix Table D6), the channel takes its market form and is the clearest welfare effect. Daily caloric intake per adult equivalent fell by 16 percent in treated areas relative to less-affected districts (Table 4). The triple difference assigns non-herders a 13 percent decline and herders a further, imprecisely estimated 8-point differential, for an implied herder total of 21 percent ($p < 0.05$). Food insecurity moves in step: the probability that a non-herder household reports any of the five FIES items rose by 7.9 percentage points from a 2022 control-group base of 10.0 percent, an increase of roughly four-fifths.

The item-level pattern shows more than a dietary adjustment. The increases concentrate on the binding margins: among non-herders, "hungry but did not eat" rose by 4.8 percentage points from a 3.5 percent base (Table 4), and "ran out of food" by 5.4 points (Appendix Table D2). The "ran out of food" margin rose significantly more for non-herders than for herders (triple difference -0.044, $p < 0.05$); the remaining herder differentials are also negative but imprecise. The asymmetry fits how each group absorbs the shock: herders can slaughter from their own herd for meat even as total calories fall and the asset base erodes, while non-herders buy their food and stand exposed to cash and price constraints.

of selling prices "reduced to half." The annual HSES recall, dominated by autumn sales after prices recovered, averages this transient out. Authors' calculations from the NSO administrative aimag-month wether-sheep price series; replication package.

[30] Control herders sold about 40 percent more livestock as their herds fell roughly 10 percent; in the worst-hit soums the herd decline was far larger and far exceeded what sales account for (Section 4; authors' calculations from the HSES livestock module).

Table 4: Caloric Intake and Food Insecurity, by Herder Status

	Caloric intake		Food insecurity (any)		Hungry, did not eat	
	DD	TD	DD	TD	DD	TD
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	-0.160***	-0.130**	0.054*	0.079**	0.040***	0.048***
	(0.055)	(0.051)	(0.028)	(0.037)	(0.015)	(0.018)
Treated × Post × Herder		-0.076		-0.063		-0.018
		(0.089)		(0.051)		(0.018)
Observations	4,254	4,254	4,130	4,130	4,228	4,228
Control mean, 2022	7.677		0.075		0.025	
Herders	7.668		0.038		0.010	
Non-herders	7.682		0.100		0.035	

Notes: Each column pair reports one outcome: daily caloric intake (log kcal per adult equivalent), an indicator for reporting any of the five FIES food-insecurity items, and the single FIES item "hungry but did not eat." DD columns report the difference-in-differences; in the TD columns, Treated × Post is the effect for non-herders and Treated × Post × Herder the additional effect for herders. Specification, sample, and weights as in Table 2. The bottom rows report population-weighted 2022 control-group means, overall and by baseline herder status; the caloric-intake mean is in log points (about 2,160 kcal per adult equivalent per day). Item-by-item estimates appear in Appendix Table D2. Standard errors, clustered at the baseline soum level, in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Two forces made poorer non-herders vulnerable, and the first is a national price shock. The prices households actually paid for staples, measured as spending per unit purchased (unit values), rose steeply between 2022 and 2024: flour by 54 percent, bread by 50 percent, milk by 49 percent, and mutton by 19 percent (Appendix Table D8).[31] Because the increase was national rather than specific to treated soums, the difference-in-differences nets it out rather than estimates it. Prices in fact evolved almost identically across treated and control soums, including for meat, the largest food item in poor households' baskets.[32]

The second force is the loss of the buffer that normally blunts food prices: own production. Among poor treated non-herders, the value of own-produced food fell by roughly a third between 2022 and 2024, while it rose 7 percent among their counterparts in less-affected districts, and their agricultural sales moved the same way (Appendix

[31] Part of a measured unit-value rise can reflect substitution toward cheaper items or qualities rather than pure price inflation.

[32] The treated–control difference on meat prices is statistically zero (unit-value DD: mutton +0.009, beef −0.017, pooled meat +0.001 log points; NSO administrative aimag-month prices show the same null), and across the full NSO food basket the treated–control gap in 2022–2024 food inflation is about one percentage point (authors' calculations from HSES unit values and NSO administrative prices; replication package). Mutton, the largest meat item among the poor, rose nationally from 9,727 to 11,540 MNT/kg (Appendix Table D8).

Tables D8 and D6). With only 77 treated households in this cell, each component is individually imprecise; the direction is nonetheless consistent across own-production and sales and aligns with the caloric and food-insecurity estimates above. The broadest measure of non-market food access, the value of food consumed from own production plus food received as gifts, measured in the food-consumption module rather than the income module, does fall significantly for poor treated non-herders (Appendix Table D7), consistent with the loss landing on subsistence access as a whole rather than on any single component. Poor households also spent 47 percent of their budgets on food, against 29 percent for better-off households (Appendix Table D8), so the same national inflation bought less protection precisely where budgets were tightest. Food inflation mattered, in short, because it met reduced subsistence production and high food shares in the same households.

This mechanism ties the food-security and quantile results together. Non-herder food losses are largest at the bottom of the distribution, with a further significant loss near the top (Appendix Table E2), and caloric intake falls despite price pressure common to both groups. The non-herder effect is a real reduction in food consumed by poorer treated households, not higher food spending or a mechanical price artifact.

5.4 Does Insurance Help?

If the poverty effect runs through livestock, insurance purchased before the winter should cushion it, and the split-sample evidence points that way. Among baseline herders, the poverty effect is 0.407 for those without livestock insurance or tax payments, our proxy for index-based livestock insurance (Bertram-Hümmer and Kraehnert, 2018), and a statistically insignificant 0.123 for those with such payments (Appendix Table G1). The formal test is the Treated $\times$ Post $\times$ Insured interaction, which is -0.135 against an uninsured-herder effect of 0.362 in the same model; taken at face value, the point estimate implies that insurance offsets just over a third of the herder poverty effect. The estimate cannot bear that much weight on its own. Its 95 percent confidence interval, roughly $[-0.46, 0.19]$, excludes neither zero nor a full offset; the insured cell is spread over only 91 soum clusters; insured herders were not spared in the pooled model; and insured herders were wealthier at baseline, so part of any gap reflects selection. We therefore read insurance as suggestively protective rather than as a demonstrated buffer, a pattern consistent with the asset-channel interpretation and with prior evidence on index-based livestock insurance (Mogge and Kraehnert, 2025), not evidence that the existing product works.

Reach would bind even if the product worked. Our proxy classifies 45 percent of herder-year observations as insured (568 of 1,262), but it counts livestock-tax payments alongside genuine index-insurance premiums and therefore overstates true coverage; given the baseline wealth gradient in who holds it, even a fully effective product would have left much of the exposed population, and its poorer part in particular, unprotected. Program history bears this out: annual take-up ranged between 8.5 and 15.5 percent of herders in covered provinces, falling to about 7 percent by 2015, and the subsidized layer created for poor households was discontinued when even a small fee proved too high a hurdle (World Bank, 2016).

These estimates nonetheless speak to a live design question in Mongolia's dzud response: whether scarce resources should flow to ex-ante risk transfer, anticipatory assistance, or ex-post relief. The anticipatory margin has experimental support in this setting: pre-winter cash raised post-dzud herd sizes among small-herd pastoralists (Mogge et al., 2025). Our evidence cannot rank these instruments, but it identifies whom each must reach: exposed herders before mortality is realized, and already-poor non-herders whose losses run through food access rather than herds (Appendix Table D7).

5.5 How Robust Are the Results?

The central identification concern is that treated and control soums were already on different welfare paths, and honest confidence intervals (Rambachan and Roth, 2023), calibrated to the pooled cross-section pre-trends, put a number on how much divergence the results tolerate (Appendix C).[33] The poverty effect survives a post-period violation up to 1.25 times the largest observed pre-period deviation, and the consumption effect up to 1.0 times. The income effect does not survive at all: its breakdown value is zero. We therefore treat the income decomposition shares as indicative (Section 5.3) and rest

[33] The panel itself has no pre-period, so the exercise is a hybrid: pre-trend coefficients and their variance block come from pooled cross-section event studies on the panel soums, the post-period entry is the panel estimate with its own variance, and the cross-covariance between the two blocks is set to zero as an independent-samples approximation. Appendix C details the construction; the bounds are best read as measuring how much differential trending the design tolerates rather than as exact confidence sets for the panel estimate.

the asset-destruction mechanism on the physical herd evidence instead. With a single pre-period, these bounds are informative rather than decisive.

The triple difference supplies a check that does not depend on cross-soum trends. It compares herders with non-herders inside the same soum, so any area-level shock common to both groups differences out. To explain away the herder result, an omitted shock would need to strike herders far harder than their non-herder neighbors within the same treated soum; given the timing and the mechanism, livestock exposure is the most direct candidate for that differential.

Selection on unobservables and placebo outcomes point the same way. Adding controls moves the poverty DD away from zero, from 0.126 to 0.151 in the first-differenced specification the stability test uses, while the herder triple difference is essentially unchanged (Appendix Table F3); selection on unobservables would therefore need to be far stronger than selection on observables to eliminate the poverty result.[34] Falsification tests find small and insignificant effects for groups without livestock exposure, for slow-moving assets, and in pre-dzud cross-sections (Appendix Figure F1).

Neither the estimator nor the treatment cutoff carries the result. The poverty DD is positive and significant under OLS (0.126), IPW (0.121) (Abadie, 2005), and doubly robust DD (0.079) (Sant'Anna and Zhao, 2020), and the herder triple difference stays between 0.19 and 0.21 across OLS, fixed effects, and IPW (Appendix Table F2).[35] Estimates are consistent in sign and pattern across the climate-based treatment definitions, attenuating at looser thresholds as a severity gradient predicts; definitions built from self-reported losses appear alongside as a descriptive benchmark rather than a causal design (Appendix F.5). Dropping the middle of the severity distribution entirely, so that only the most extreme soums face the least affected, leaves both coefficients essentially unchanged (poverty DD 0.145, herder differential 0.199; Appendix Table F5).

[34] The Oster (2019) statistic for the poverty DD is $\delta = -17.1$: selection on unobservables would need to be about seventeen times as strong as selection on observables to drive it to zero, assuming $R_{\text{max}} = 1.3R_{\text{long}}$. For the herder triple difference, the coefficient is essentially unchanged when controls are added, making the implied statistic numerically unstable and very large in magnitude; we interpret this as evidence that observables explain little of the herder differential rather than as a precise point estimate.

[35] A doubly robust triple difference is not directly comparable to these estimates. The estimator in Sant'Anna and Zhao (2020) is defined for a single treated–control contrast, so the triple difference must be assembled from separate covariate-adjusted DDs for herders and non-herders. Because each subsample conditions on baseline livestock holdings and agricultural dependence, the very channel through which the dzud operates, the resulting contrast estimates Caron's covariate-adjusted differential rather than the raw differential we target (Caron, 2025).

Measurement choices matter as little. Rescaling the national poverty line by twenty percent in either direction leaves the DD positive and significant, at 0.130 and 0.139 (Appendix Table F1). Across weighting schemes the poverty DD stays significant at 0.10 to 0.15, with only the unweighted triple difference losing significance (Appendix F.2). Sample restrictions work in the same direction: excluding households interviewed in the first quarter, or heads resident in the soum for fewer than five years, leaves the estimates intact or larger.

Geographic checks support the same interpretation. Dropping the five Western aimags leaves the poverty DD unchanged at 0.152 and raises the herder differential to 0.232, while adding Ulaanbaatar as an urban control attenuates the area-level DD to 0.109 but moves the triple difference only from 0.206 to 0.220 (Appendix Tables F1, F6). No single estimator, treatment cutoff, poverty line, weighting choice, or geographic sample drives the core finding.

The geographic concentration of this event nonetheless bounds what the area-level estimate can claim. Half of treated households live in the Eastern region (Figure 1), so the DD is identified largely from Eastern soums. The herder differential is the more portable parameter: it is stable when the Western aimags are dropped and when Ulaanbaatar is added as a control, and because past dzuds have struck different regions, a future event with a different footprint should reproduce the within-soum incidence even if the area-level magnitude differs.

6 Conclusion

A country can lose one in every eight of its livestock and still record a steep fall in poverty; that arithmetic is why the cost of Mongolia's 2023/24 dzud went unmeasured. Using the first nationally representative household panel to bracket a major dzud, we estimate that the shock erased 15.2 percentage points of poverty reduction in the most-affected soums while poverty was falling rapidly everywhere else, and that total consumption grew about 14 percent less than it would have without the shock. The cost fell overwhelmingly on herders and reached poorer non-herders through food.

The two populations were hit through different routes. For herders, the mechanism is asset destruction. Their herds ended the panel about a third smaller on net, and gross winter losses reached near half of the baseline herd, partly offset by restocking. The income loss ran overwhelmingly through livestock income, a quantity channel rather than a sustained price channel: animals died before they could generate wool, cashmere, milk, meat, or sale income, and the herder food loss traveled with the herd, as own-produced meat and dairy disappeared alongside cash income. Non-herders, by contrast, cut real food consumption, and their food insecurity rose by roughly four-fifths from a baseline of about one in ten households, a pattern consistent with lost subsistence production reducing their buffer against rising food prices. In both groups the losses concentrate low in the welfare distribution, which places part of the dzud's burden outside the herding population where prior evidence had located it (Groppo and Kraehnert, 2016, 2017).

These findings point to four policy margins. First, support needs to arrive before the winter shock is fully realized: the poverty effect is a threshold-crossing effect among vulnerable herders, so assistance delivered only after animals have died may miss the moment when poverty entry can still be prevented, and a recent Mongolian trial shows that anticipatory cash transfers raised post-dzud herd sizes among small-herd pastoralists (Mogge et al., 2025). Second, livestock insurance is a candidate instrument for asset protection: our estimates are consistent with a protective role, especially among uninsured herders, but they are too imprecise to establish effectiveness, and the coverage that exists skews toward wealthier herders. Third, policy should not focus only on herd owners, since poor non-herders were affected through food access (Appendix Table D7), which suggests a role for food-price-sensitive transfers and support for subsistence production. Fourth, recovery requires attention to herd rebuilding, because the loss of livestock stock reduces future income, own-consumption, and resilience.

Several limitations warrant acknowledgment. First, the estimates capture short-run effects of an unusually severe event, so the medium-run recovery path remains open. Second, the estimates are likely conservative: control soums also experienced some winter stress, so the ATT captures incremental severity rather than the full dzud effect; government-led anticipatory response programs, complemented by UN, IFRC, and INGO partners, partially cushioned the blow; and the survey's rolling interview design attenuates the measured effect for households interviewed early in 2024. Each of these features biases the difference-in-differences toward zero. Third, the estimates capture contemporaneous welfare losses but not the deferred cost of a smaller surviving herd, and therefore understate the full cost of the event: a herd left about a third smaller reduces future sales, wool and cashmere production, milk and meat consumption, and collateral value. Characterizing the medium- and long-term trajectory of dzud-affected herders, including whether herds recover, herders permanently exit pastoralism, or income shifts toward wage labor, is a natural task for subsequent rounds of the household survey.

The paradox that opens this paper has a plain resolution. The dzud left no mark on the national statistics because its cost took the form of poverty reduction that never arrived, concentrated on the households whose wealth stood in the path of the winter while the rest of the country moved forward. Three catastrophic dzuds in twenty-five years make this a recurrent risk in Mongolia, not an exceptional loss, and the world's cold rangelands share the exposure. That climate adaptation is poverty policy is by now a central theme of the climate-poverty literature (Hallegatte et al., 2016, 2017); our results show how immediate the link becomes in pastoral economies. The choice is whether to respond after herds have died and poverty gains have been reversed, or to insure and support exposed households before the next winter arrives.

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A Variable Definitions

Table A1: Variable Definitions

Variable	Definition	Source
Panel A: Outcome Variables		
Poverty rate	= 1 if real consumption PAE falls below the national poverty line (16,882 MNT/day in 2024; 13,744 in 2022)	HSES; NSO poverty line
FGT poverty measures	Foster–Greer–Thorbecke indices (Foster et al., 1984): FGT_0 = headcount (share below the poverty line); FGT_1 = poverty gap (average shortfall below the line, as a share of the line); FGT_2 = severity (squared shortfalls, weighting the poorest most)	HSES (derived)
Log income (PAE)	Natural log of real total household income per adult equivalent, deflated to 2024 constant prices. Uses non-negative total income (negative farm/enterprise floored to zero)	HSES
UQTE	Unconditional quantile treatment effects on consumption at welfare-anchored quantiles (Q15, Q30, Q50, Q70, Q90)	HSES
Panel B: Placebo Outcomes (Falsification)		
Non-ger dwelling	= 1 if household dwelling is not a ger (yurt); slow-moving asset unaffected by a single-season shock	HSES
Owns house / land	= 1 if household owns their house (respectively, land); slow-moving assets	HSES
Panel C: Heterogeneity Variables		

Continued on next page

Table A1 continued from previous page

Variable	Definition	Source
Herder (baseline)	= 1 if the household head is classified as a herder from the 2022 HSES employment module and livestock roster. A head qualifies through either of two routes. *Occupation* (which classifies most herders): the head's main job is in animal production (HSES industry code 114), the head reports agricultural work whose purpose is herding, or the head is an employer or own-account worker in a livestock-owning household. *Livestock ownership*: a household owning at least 50 head with no member employed elsewhere has its head, or failing that its eldest working-age member, classified as a herder. Capital-city residents are never classified as herders; in aimag centres a 50-head minimum also applies to the occupation route. Fixed at baseline (2022) to avoid endogeneity from post-dzud occupation switching	HSES 2022 employment module; livestock roster
Insurance (baseline)	= 1 if household reported positive expenditure on livestock insurance or tax at baseline	HSES 2022
Wage-primary	= 1 if baseline wage income share > 70%; used in falsification tests to identify households with no livestock channel	HSES
WB welfare class (baseline, 3-class)	Categorical: Poor (consumption PAE below the national poverty line of 16,882 MNT/day, 2024 prices), Vulnerable (16,882–27,142 MNT/day), Middle-Upper (≥27,142 MNT/day). The 5-class scheme (Poor, Vulnerable, Emerging-middle, Secure-middle, Upper) is collapsed to 3 classes because Secure-middle and Upper are sparse in the herder panel. Used in the non-herder income decomposition (Table D6, Panel B) and the food-exposure analysis (Table D8)	HSES (derived)
Panel D: Household Controls		

Continued on next page

Table A1 continued from previous page

Variable	Definition	Source
Region	Four regions: West, Highlands, Central, and East. Interacted with Post in preferred specification to absorb region-specific trends	HSES
Quarter dummies	Interview quarter indicators (Q1–Q4) for seasonality	HSES
Asset resale value (IHS, baseline)	Inverse hyperbolic sine of total durable asset resale value at baseline; used as a continuous wealth control	HSES 2022
Livestock holdings (IHS, baseline)	Inverse hyperbolic sine of baseline livestock in bod units; used as a continuous wealth control	HSES 2022; NSO
Panel E: Sample and Weight Variables		
Population weight	Household sampling weight × household size, adjusted for panel attrition; primary weight throughout	HSES
Household weight	Household-level sampling weight, adjusted for panel attrition	HSES
Baseline soum	Soum of residence in 2022; used for treatment assignment, SE clustering, and fixed effects	HSES 2022
Panel F: Livestock-Sale Decomposition Variables		
bod sold (per HH)	bod-standardized count of live animals sold during the year. Used in the continuing-seller log decomposition (Table D4 Panel B) and the extensive-margin exit summary (Table D5)	HSES livestock module
Per-bod price (implied)	Livestock-sales revenue / bod sold, in 000s MNT per bod; computed only for HHs with positive sales. Used in the continuing-seller log decomposition; the implied price is interpreted as an HH-level approximate average producer price for the year	HSES (derived)
Share selling positive bod	Population-weighted share of baseline-herder households with positive bod sold in the year; used in the extensive-margin exit summary (Table D5)	HSES (derived)
Panel G: Migration / Stayer Variables		

Continued on next page

Table A1 continued from previous page

Variable	Definition	Source
Long-term stayer (5/10 yr)	= 1 if household head resided in current soum for 5+ (primary) or 10+ (strict) years at baseline (2022)	HSES 2022 migration module

Notes: PAE = per adult equivalent (modified OECD scale: 0.3+0.7×adults+0.5×children). MNT = Mongolian Tugrik. UB = Ulaanbaatar (excluded). bod = standardized large-livestock-equivalent unit (1 horse or cow = 1 bod; 1 camel = 1.5 bod; 6 sheep = 1 bod; 8 goats = 1 bod). IHS = inverse hyperbolic sine. All monetary values in 2024 constant prices unless noted. Controls interacted with Post in regressions. See Appendix F.5 for alternative treatment definitions.

B Sample, Balance, and Attrition

Table B1: Sample Construction and Treatment Assignment

Panel A: Sample construction	Soums	Households	Household-years
Full HSES 2022 baseline (excl. Ulaanbaatar)	330	18,318	—
Balanced panel (observed 2022 & 2024)	175	2,128	4,256

Panel B: Treatment assignment	Treated	Control	Total
Soums			
All non-UB soums	79	251	330
Panel soums	46	129	175
Households (baseline 2022)			
All households	530	1,598	2,128
Herder	153	478	631
Non-herder	377	1,120	1,497
Household-year observations	1,060	3,196	4,256

Notes: Panel A shows sample construction from the full HSES 2022 baseline (excluding Ulaanbaatar, where <1% of households engage in pastoral herding); the balanced panel includes households observed in both 2022 (pre-dzud) and 2024 (post-dzud). Panel B shows treatment assignment: a soum is treated if its 2023/24 winter temperature deviation or snow cover anomaly exceeded the 85th percentile (P85) of the cross-soum distribution among all 330 non-UB soums (Section 3.1 gives the formal definition). "All non-UB soums" is the full treatment universe; "Panel soums" the subset containing HSES panel households. Herder status is measured at 2022 baseline to avoid endogeneity from post-dzud occupation switching; attrition is orthogonal to treatment (Appendix B.1). Source: HSES 2022–2024 panel, ERA5 temperature reanalysis, MODIS snow cover.

Table B2: Covariate and Outcome Balance at Baseline (2022)

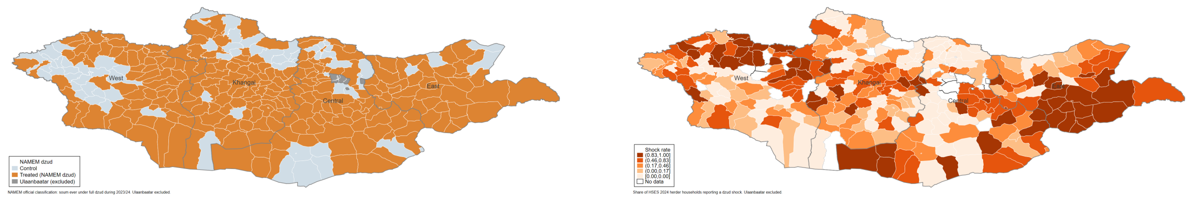
	Control (N=1,598)	Treated (N=530)	Raw Diff.	Norm. Diff
Panel A: Baseline Outcomes (2022)				
Poverty rate	0.335	0.234	-0.101**	-0.225
Log consumption (PAE)	9.938	10.013	0.074	0.172
Log income (PAE)	15.848	15.947	0.099*	0.169
Log food consumption (PAE)	8.886	9.016	0.130***	**0.333**
Panel B: Predetermined Covariates				
Head age (years)	44.7 (13.0)	44.1 (12.4)	-0.5	-0.042
Has female head	0.141	0.160	0.019	0.053
Household size	4.1 (1.7)	4.2 (1.7)	0.0	0.018
Head completed secondary	0.373	0.370	-0.003	-0.006
Head completed tertiary	0.177	0.157	-0.020	-0.053
Urban household	0.384	0.430	0.045	0.092
Panel C: Livelihood Characteristics				
Has herder head	0.402	0.392	-0.010	-0.020
IHS asset resale value	17.48 (1.41)	17.50 (1.19)	0.02	0.013
IHS livestock (bod)	2.51 (2.39)	2.68 (2.62)	0.16	0.066
Has livestock insurance	0.192	0.304	0.112	**0.262**
Members working (n)	0.7 (0.8)	0.8 (0.8)	0.1	0.075
Wage income share	29.216	30.388	1.172	0.034
Agricultural income share	28.142	30.313	2.171	0.063
Business income share	3.407	2.518	-0.889	-0.068
Non-labor income share	38.525	35.669	-2.856	-0.096
Region (population shares):				
Western	0.311	0.000	-0.311***	**-0.950**
Khangai Highlands	0.396	0.144	-0.252**	**-0.591**
Central	0.248	0.356	0.108	0.238
Eastern	0.045	0.500	0.455***	**1.186**
Panel D: Covariate Changes (Δ 2022–2024)				
Predetermined:				
Δ Household size	0.136	0.074	-0.062	-0.050
Δ Urban	-0.007	-0.008	-0.000	-0.004
Δ Herder status	-0.011	-0.012	-0.000	-0.001
Livelihood (potential mechanisms):				
Δ Insurance	0.068	-0.096	-0.165***	**-0.348**
Δ IHS livestock	-0.070	-0.266	-0.196	-0.128
Δ Wage share	5.590	6.216	0.626	0.022
Δ Agricultural share	-2.156	-5.054	-2.898*	-0.134
Δ Non-labor share	-4.016	-1.801	2.215	0.087
N (households)	1,598	530		

Notes: Baseline (2022) population-weighted values. Cells are `mean (SD)` for continuous variables and the share for binaries. *Raw Diff.* $= \bar{x}_1 - \bar{x}_0$ (Treated − Control, population-weighted); stars from a population-weighted regression of each variable on the treatment indicator, SEs clustered at baseline soum. Normalized difference (*Norm. Diff*) $= (\bar{x}_1 - \bar{x}_0)/\sqrt{(s_1^2 + s_0^2)/2}$ (Imbens and Rubin, 2015); **bold** $= |d| \geq 0.25$. PAE = per adult equivalent (OECD-modified scale); bod = large-livestock-equivalent unit (Appendix A). Predetermined covariates (Panel B, incl. region) match the IPW/DR-DD propensity-score specification; Panel D shows balance in 2022–2024 changes of time-varying covariates. * p<0.10, ** p<0.05, *** p<0.01.

Table B3: Balance in Interview Timing by Treatment Status

	Control (N=1598)	Treated (N=530)	Norm. Diff
Panel A: Interview Quarter (2022)			
Q1 (Jan–Mar)	0.481	0.539	0.116
Q2 (Apr–Jun)	0.290	0.244	-0.104
Q3 (Jul–Sep)	0.139	0.164	0.070
Q4 (Oct–Dec)	0.090	0.053	-0.144
Panel B: Interview Quarter (2024)			
Q1 (Jan–Mar)	0.289	0.180	**-0.259**
Q2 (Apr–Jun)	0.216	0.300	0.193
Q3 (Jul–Sep)	0.236	0.313	0.173
Q4 (Oct–Dec)	0.259	0.206	-0.126

Notes: Population-weighted interview-quarter distributions by treatment status: Panel A is the 2022 wave, Panel B the 2024 wave. Normalized difference (Imbens and Rubin, 2015); **bold** = $|d| \geq 0.25$.

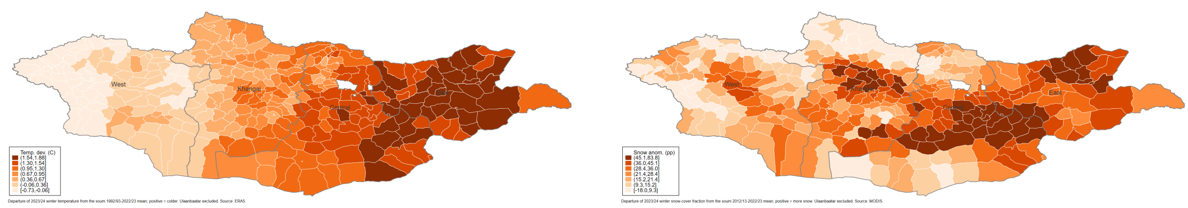


(a) NAMEM Ever-Dzud Classification

(b) Self-Reported Dzud Shock Rate

Figure B1: Official Dzud Classification and Self-Reported Shocks

Notes: Panel (a) shows the NAMEM official dzud classification (257 of 330 non-Ulaanbaatar soums classified as full dzud at least once during winter 2023/24); the classification flags the presence of dzud-type conditions and is deliberately sensitive for early warning, so its breadth reflects hazard-type detection rather than uniform severity. Panel (b) shows the share of herder households in the HSES 2024 cross-section who reported a dzud shock, binned into five quantiles. The herder-only restriction reflects the dzud's livestock-based transmission channel; non-herder responses capture area spillovers rather than direct livestock exposure. Both maps exclude Ulaanbaatar. The preferred treatment and the livestock-loss validation are in Figure 1.



(a) Winter temperature deviation (°C)

(b) Winter snow-cover anomaly (pp)

Figure B2: Continuous Winter-Severity Measures by Soum, 2023/24

Notes: The two continuous severity measures underlying the binary treatment (darker = more severe): Panel (a) the winter temperature deviation from the soum's 1992/93–2022/23 mean (positive = colder than normal, ERA5), Panel (b) the winter snow-cover anomaly from the 2012/13–2022/23 mean (positive = more snow, MODIS). A soum is treated when either measure exceeds its 85th percentile (P85) across the 330 non-Ulaanbaatar soums shown (Eq. 3).

B.1 Attrition and Representativeness

Normalized differences between the balanced panel and the full 2022 baseline sample never exceed 0.06 in absolute value, far below the 0.25 threshold, so panel attrition leaves the analysis sample representative of the baseline population on outcomes, covariates, and herder status (Table B4).

Table B4: Attrition Balance: Panel vs. Full Sample at Baseline (2022)

	Panel	Full Sample	Raw Diff	*p*-value	Norm. Diff
Panel A: Baseline Outcomes					
Poverty rate	0.257	0.265	-0.008	0.401	-0.017
Log consumption (PAE)	9.820	9.805	0.015	0.103	0.033
Log income (PAE)	15.662	15.671	-0.010	0.421	-0.017
Log food consumption (PAE)	8.748	8.726	0.022	0.007	0.056
Panel B: Predetermined Covariates					
Head age (years)	48.746	48.577	0.169	0.564	0.012
Female head	0.228	0.238	-0.010	0.234	-0.024
Household size	3.421	3.373	0.047	0.189	0.027
Secondary education	0.159	0.154	0.005	0.492	0.014
Tertiary education	0.177	0.182	-0.005	0.502	-0.014
Urban	0.435	0.414	0.020	0.042	0.041
Panel C: Livelihood Characteristics					
Herder head	0.297	0.294	0.003	0.752	0.006
N	2,128	18,318			

Notes: Baseline (2022) values, excl. Ulaanbaatar. Panel = 2,128 households observed in both 2022 and 2024; Full Sample = all 18,318 baseline households. Normalized difference (Imbens and Rubin, 2015); **bold** = $|d| \geq 0.25$.

C Pre-Trend Evidence and Sensitivity to Parallel Trends Violations

This appendix collects the design-validation evidence behind the parallel-trends and parallel-gap-trends discussion in Section 4.2: first the pre-trend figures, then the Honest-DiD sensitivity analysis, which asks how large a violation of parallel trends would need to be to overturn the estimates. The pre-trend figures use pooled cross-sections restricted to the 175 panel soums; the panel DD/TD uses the balanced 2022–2024 panel.

Pre-Trend Evidence

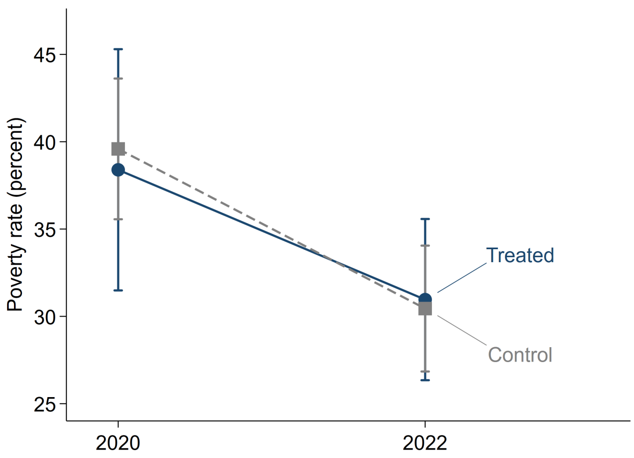


Figure C1: Poverty Fell in Parallel in Treated and Control Soums Before the Dzud

Notes: Each point is the population-weighted poverty rate among households in treated or control soums in that year's HSES cross-section, restricted to the 175 panel soums; bars are 95 percent confidence intervals, clustered at the baseline soum. The two groups decline in parallel over 2020–2022, supporting the parallel-trends assumption. The treatment-period comparison is identified within panel households and is shown in Figure 2.

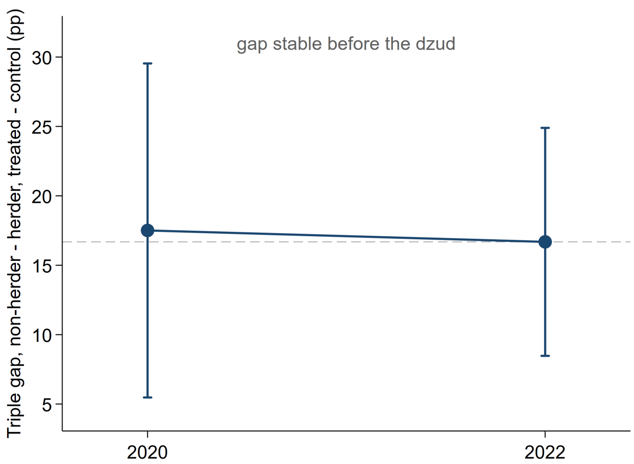


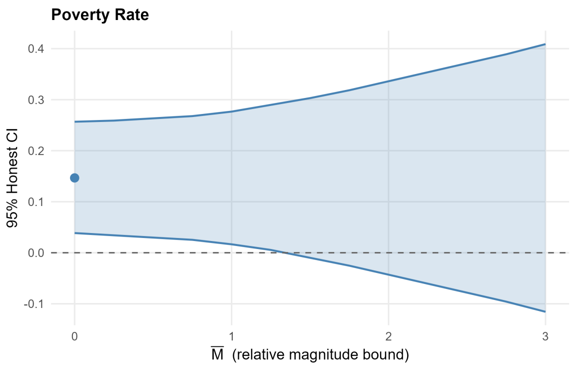
Figure C2: The Herder–Non-Herder Poverty Gap Was Stable Before the Dzud

Notes: Each point is the difference between two gaps: the non-herder-minus-herder poverty gap in treated soums minus the same gap in control soums, from HSES cross-sections restricted to the 175 panel soums; bars are 95 percent confidence intervals, clustered at the baseline soum. The dashed line marks the 2022 level: identification requires only that this gap is stable before the dzud, and the 2020 and 2022 points are nearly identical. The treatment-period effect is the panel triple difference reported in Table 2.

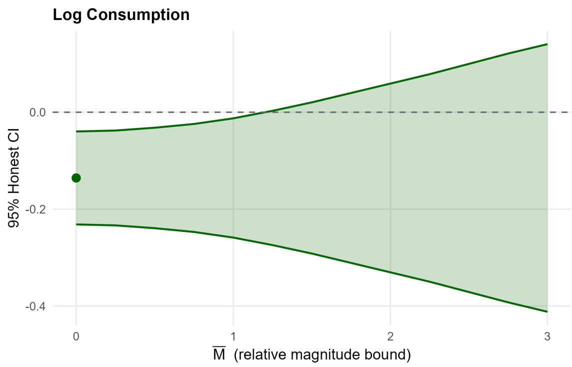
Sensitivity Analysis

Figure C3 reports the honest confidence intervals for the poverty, consumption, and income DDs, following Rambachan and Roth (2023).

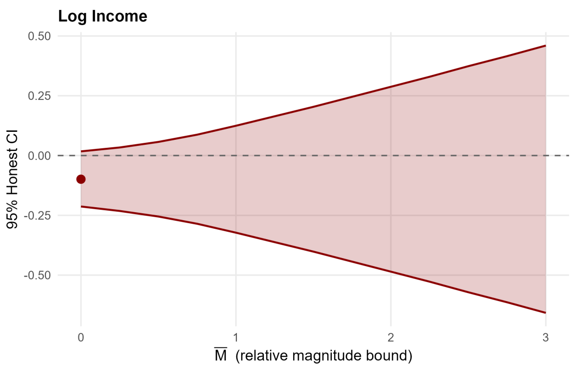
The exercise is a hybrid by necessity: the panel has no pre-period, so the pre-trend coefficients and their variance-covariance block come from the pooled cross-section event studies restricted to the 175 panel soums, while the post-period entry is the panel DD estimate with controls and its own variance. Because the two blocks are estimated on different samples, their cross-covariance is set to zero; this is an approximation, since the 2022 wave contributes households to both estimates. The bounds therefore combine cross-section pre-trend information with the panel treatment effect, and we read them as informative about how much differential trending the design tolerates rather than as exact confidence sets for the panel estimate.



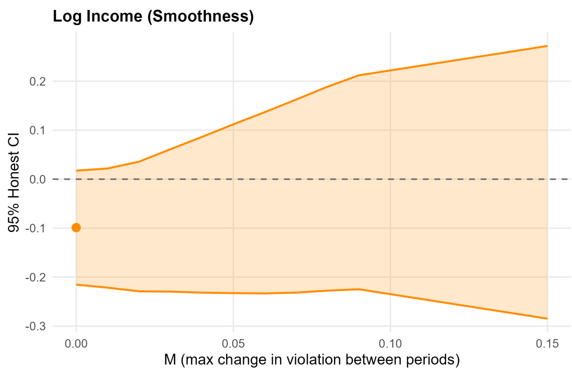
(a) Poverty (Relative Magnitudes)



(b) Log Consumption (Relative Magnitudes)



(c) Log Income (Relative Magnitudes)



(d) Log Income (Smoothness)

Figure C3: Sensitivity Analysis: Honest Confidence Intervals

Notes: Honest 95% confidence intervals following Rambachan and Roth (2023). The relative-magnitudes restriction ($\Delta^{\mathrm{RM}}(\bar{M})$, panels a–c) bounds the post-treatment parallel-trends violation at $\bar{M}$ times the largest pre-treatment violation; the smoothness restriction ($\Delta^{\mathrm{SD}}(M)$, panel d, income only) bounds the per-period change at M and requires $\geq$2 pre-periods. Pre-trend coefficients are calibrated from pooled cross-section event studies restricted to the 175 panel soums; post-treatment estimates from the panel DD with controls. The dot at $\bar{M} = 0$ is the original DD estimate. Breakdown values: poverty $\bar{M}^* = 1.25$, total consumption $\bar{M}^* = 1.0$, income $\bar{M}^* = 0$ and $M^*_{\mathrm{SD}} = 0$ (the income DD does not survive any parallel-trends violation).

D Outcome Regressions and Decompositions

This appendix reports the regressions behind the compact body table (Table 2) and the income and consumption decompositions behind the two channels of Section 5.3. Table D1 reports the difference-in-differences (DD) and triple-difference (TD) coefficients for the three log outcomes (income, total consumption, and food consumption per adult equivalent); the full poverty specification, with all main effects and the OLS level differences, is available in the replication package. Table D3 decomposes total income into grouped components and estimates separate DDs by subgroup; Tables D4 and D5 detail the herder livestock channel, and Tables D6 and D8 the non-herder side.

Table D1: DD and TD Coefficients Across Outcomes

	Log income	Log cons.	Log food
Panel A: Difference-in-differences			
Treated × Post	-0.113*	-0.141***	-0.210***
	(0.060)	(0.048)	(0.075)
Panel B: Triple difference			
Treated × Post	-0.053	-0.088*	-0.183***
	(0.063)	(0.047)	(0.063)
Treated × Post × Herder	-0.151	-0.131*	-0.067
	(0.101)	(0.078)	(0.120)
Controls × Post	Yes	Yes	Yes
Household + Year FE	Yes	Yes	Yes
Observations	4,246	4,254	4,254
Control mean, 2022	766,602	685,372	232,131
Herders	728,909	581,829	234,805
Non-herders	791,949	755,001	230,332

Notes: Panel A reports the pooled (area-level) difference-in-differences, Treated × Post. Panel B reports the triple difference: Treated × Post is the non-herder difference-in-differences and Treated × Post × Herder the additional herder effect, i.e. the herder–non-herder differential (the herder effect is their sum). All use the preferred covariate-adjusted TWFE specification (household and year fixed effects, baseline covariates and region interacted with post; as in Table 2). Income, consumption, and food are in logs per adult equivalent (2024 constant prices); the poverty headcount is in the body (Table 2). The lower panel reports the population-weighted 2022 control-group level of each outcome, in MNT per adult equivalent per month (income, consumption, and food are on the same real spatio-temporal price basis, deflated by the soum-level price index), overall and split by baseline herder status. Identification confidence is outcome-specific: consumption withstands larger parallel-trends violations than income, the least robust (Appendix C). SEs clustered at baseline soum in parentheses; population weights. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table D2 reports the full item-by-item food-insecurity breakdown and the triple difference. Among the FIES items, the treated–control increases are most precise for “hungry but did not eat” (Q7) and “ran out of food” (Q6); “whole day without eating” (Q8) is smallest and insignificant. The triple difference on “ran out of food” is negative and significant (-0.044, $p < 0.05$).

Table D2: Caloric Intake and Food-Insecurity Items: DD and Triple Difference

		FIES food-insecurity items (=1 if yes)					
	Caloric intake	Worried about food	Skipped a meal	Ran out of food	Hungry, did not eat	Whole day without eating	Any FIES item
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Difference-in-differences							
Treated × Post	-0.160***	0.042*	0.034**	0.037**	0.040***	0.018	0.054*
	(0.055)	(0.024)	(0.017)	(0.015)	(0.015)	(0.013)	(0.028)
Observations	4,254	4,216	4,212	4,194	4,228	4,226	4,130
Panel B: Triple difference							
Treated × Post	-0.130**	0.069**	0.043*	0.054***	0.048***	0.031*	0.079**
	(0.051)	(0.030)	(0.024)	(0.020)	(0.018)	(0.019)	(0.037)
Treated × Post × Herder	-0.076	-0.068	-0.025	-0.044**	-0.018	-0.033	-0.063
	(0.089)	(0.041)	(0.032)	(0.022)	(0.018)	(0.024)	(0.051)
Observations	4,254	4,216	4,212	4,194	4,228	4,226	4,130
Control mean, 2022	7.677	0.067	0.022	0.024	0.025	0.018	0.075
Herders	7.668	0.037	0.007	0.006	0.010	0.005	0.038
Non-herders	7.682	0.087	0.032	0.036	0.035	0.026	0.100

Notes: Each column is a separate outcome: daily caloric intake (log kcal per adult equivalent) and the five FIES food-insecurity items from HSES Section 15, with the final column an indicator for reporting any item. Panel A reports the difference-in-differences; in Panel B, Treated × Post is the effect for non-herders and Treated × Post × Herder the additional effect for herders. Specification, sample, and weights as in Table 2. Column samples differ slightly because each item uses its own nonmissing observations, and the any-item indicator requires all five. The bottom rows report population-weighted 2022 control-group means, overall and by baseline herder status; column 1's mean is in log points. Standard errors, clustered at the baseline soum level, in parentheses. $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$.

Table D3: DD Income Decomposition by Component and Subgroup (Levels)

	Full Sample			Non-Herder			Herder		
	DD (000s)	SE	Share (%)	DD (000s)	SE	Share (%)	DD (000s)	SE	Share (%)
Wages	-1,925**	(942)	49.9	-2,123*	(1,212)	158.8	-1,973*	(1,101)	21.0
Livestock income	-2,801**	(1,098)	72.7	-687	(536)	51.4	-6,685**	(2,940)	71.3
Livestock sales	-1,235*	(645)		-374	(368)		-3,023*	(1,769)	
Byproduct sales	-774**	(367)		-96	(92)		-2,102*	(1,146)	
Own livestock	-332*	(174)		-120	(127)		-662*	(340)	
Own byproducts	-461	(295)		-96	(112)		-898	(695)	
Crop income	83	(83)	-2.2	15	(73)	-1.1	163	(155)	-1.7
Enterprise	718	(865)	-18.6	930	(1,009)	-69.5	-269	(223)	2.9
Nonlabor income	-23	(399)	0.6	481	(601)	-36.0	-781	(505)	8.3
Total	-3,855*	(1,987)	100.0	-1,337	(1,798)	100.0	-9,372***	(3,125)	100.0
Observations		4,256			2,994			1,262	
Memo: baseline 2022 share of total income (%), pop-weighted									
Wages			31.5			49.1			5.9
Livestock income			30.9			6.3			66.7
Crop income			0.6			0.5			0.6
Enterprise			6.9			11.5			0.2
Nonlabor income			29.9			32.6			26.0

Notes: Each row is a separate DD regression of the grouped income component (annual, 000s MNT, 2024 prices) on Treated × Post, estimated separately for the full sample, non-herder, and herder subsamples, with two-way (household, year) fixed effects and quarter × post interactions; R-squared for fixed-effects specifications is within R-squared. The components are additive by construction, and remain so under the full baseline-covariate adjustment of Table 2. Under that adjustment the total-income DD is nearly unchanged, but the livestock share of the herder income loss rises to about 83 percent (from 71.3 percent here), because covariate adjustment attenuates the smaller, noisier non-livestock components; livestock dominance is therefore robust to specification. The "Share" column is each component's share of the group's total income change (DD). The memo block at the foot reports each group's baseline 2022 composition, the share of total income from each source before the shock (population-weighted). 1 USD ≈ 3,450 MNT. SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.

Tables D4 and D5 belong together: the first decomposes the livestock-income loss, and the second shows the herd stock and market participation behind it.

Table D4: Herder Livestock-Income Decomposition

	DD	SE	Share
Panel A: Income components (DD, '000 MNT, 2024 prices)			
Sales			
Livestock sales (live animals)	$-3{,}023^{*}$	(1,769)	45%
Byproduct sales (wool, cashmere, dairy)	$-2{,}102^{*}$	(1,146)	31%
Subtotal sales	$-5{,}125$	—	*77%*
Own-consumption			
Own-consumed livestock (home slaughter)	-662^{*}	(340)	10%
Own-consumed byproducts (dairy, wool)	-898	(695)	13%
Subtotal own-consumption	$-1{,}560$	—	*23%*
Total herder livestock income	$-6{,}685^{**}$	(2,940)	**100%**
Panel B: Continuing-seller revenue: price vs. quantity ($\log R = \log P + \log Q$)			
$\log R$ – sales revenue	0.031	(0.198)	100%
$\log P$ – implied price per bod	0.183	(0.140)	594.3%
$\log Q$ – bod sold	-0.152	(0.198)	-494.3%
Identity residual ($R - P - Q$)	-0.00000	—	—
Observations (selling herder-years)			548

Notes: Baseline (2022) herder households. *Panel A* reports, for each livestock-income component, the difference-in-differences (Treated × Post; annual, 000s MNT, 2024 constant prices), its standard error, and the component's share of the total livestock-income DD; all components reflect quantity reductions from the smaller surviving herd. *Panel B* decomposes the change in livestock-sales revenue among herders who sold in both years into price and quantity using the exact log identity $\log R = \log P + \log Q$ (548 selling herder-years, balanced); the price and quantity DDs are individually insignificant, locating the revenue change in quantity rather than price. The full income decomposition (wages, crop, nonlabor, and total) is in Appendix Table D3; herd stock and market participation are in Table D5. 1 USD ≈ 3,450 MNT. DD throughout: TWFE (household, year), quarter × post; SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.

Table D5: Herder Herd Stock and Market Participation

	Herd size (bod/HH)			Selling any animals		
	2022	2024	Δ%	2022	2024	Δpp
Control	95.0	85.9	-9.7	71.6	70.9	-0.7
Treated	143.2	95.4	-33.4	86.5	76.0	-10.5
Diff-in-diff			-23.7			-9.8

Notes: Population-weighted means for baseline (2022) herder households in control versus treated soums. Herd size is in *bod* (the Mongolian large-livestock equivalent); "Selling any animals" is the share of herders with any livestock sale that year. Each block is a 2 × 2 difference-in-differences: the Δ column gives each group's own 2022–2024 change (herd size in growth-rate percent, selling in percentage points), and "Diff-in-diff" is the treated-minus-control difference in those changes. The herd-size columns report unadjusted population-weighted means; only the selling margin is covariate-adjusted (DD on the probability of any sale −5.8 pp, SE 5.2), suggestive of market exit. The net herd decline understates mortality: a residual accounting of the winter loss (baseline herd plus own-consumption and sales, less the surviving herd) implies deaths near half the baseline treated herd, partly offset by restocking. Authors' calculations from the HSES livestock module.

Table D6: Non-Herder Income Decomposition: All Non-Herders and the Baseline Poor

	BL share	Control			Treated			DD
	(%)	'22	'24	Δ%	'22	'24	Δ%	(000s)
Panel A: All non-herders								
Wages	49	12,251	18,320	50	13,131	16,924	29	-2,123*
Agric. income — sales	3	773	807	4	1,310	911	-30	-413
Agric. income — own consumption	3	828	1,060	28	880	853	-3	-259
Enterprise	11	3,342	2,670	-20	1,159	2,409	108	930
Nonlabor income	33	8,330	8,510	2	7,884	8,228	4	481
Total	100	25,545	31,394	23	24,368	29,381	21	-1,337
Households (2024)		1,120			377			
Panel B: Poor (baseline)								
Wages	37	5,907	11,817	100	6,179	11,892	92	-1,177
Agric. income — sales	4	652	884	36	1,002	359	-64	-716
Agric. income — own consumption	6	1,037	1,111	7	590	387	-34	-329
Enterprise	3	426	1,383	224	771	1,267	64	-277
Nonlabor income	50	7,942	7,933	-0	8,244	8,880	8	1,141
Total	100	15,984	23,154	45	16,768	22,805	36	-1,323
Households (2024)		269			77			

Notes: Baseline (2022) non-herder households. Panel A pools all baseline welfare classes; Panel B restricts to the baseline *Poor* (2022 consumption below the poverty line; the full classification also distinguishes *Vulnerable*, line–27,142, and *Middle-Upper*, ≥27,142 MNT/PAE/day, 2024 prices). Agricultural income is split into *cash* (livestock, byproduct, crop sales) and *in-kind* (own consumption at aimag-median prices). Cells: 2024 population-weighted means (annual, 000s MNT), within-group growth (Δ%), and the DD (Treated × Post). Stratification by the vulnerable and middle-upper classes is available in the replication package. TWFE (household, year), quarter × post; SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.

Table D7: Non-Market Food Access: Own Production and Gifts

	All non-herders (1)	Poor (2)
Own-produced food consumed	-0.623	-1.390*
	(0.393)	(0.717)
Own production + food gifts	0.013	-1.544**
	(0.521)	(0.631)
Observations	2,992	692

Notes: Difference-in-differences (Treated × Post) on the inverse hyperbolic sine of real annual household values from the HSES food module: the value of food consumed from own production, and that value plus food received as gifts (2024 prices). Column (1): all baseline (2022) non-herder households; column (2): baseline-poor non-herders (77 treated households). The income-module in-kind values in Table D6 and the food-module consumption values here measure different objects on different scales (levels in thousands of MNT there; inverse hyperbolic sine here), so magnitudes are not comparable across the two tables; for the baseline poor, both move in the same direction. Specification, sample construction, and weights as in Table 2. Standard errors, clustered at the baseline soum level, in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table D8: The Two Food Shocks Facing Non-Herder Households, by Baseline Welfare Class

Panel A: The price shock and who is exposed to it				
National change in household unit values, 2022–2024				
Item	2022 (MNT)	2024 (MNT)	Change (%)	
Red meat (mutton)	9,727	11,540	+18.6	
Flour	1,857	2,853	+53.6	
Bread	1,909	2,855	+49.6	
Milk	3,017	4,483	+48.6	
2022 food budget share, by baseline welfare class				
Poor	46.8%			
Vulnerable	37.3%			
Middle-Upper	28.6%			
Panel B: Lost own food production (MNT/HH-year, 2024 prices)				
	2022	2024	Change (%)	N
Overall (all non-herders)				
All non-herders	838.5	1,019.0	+21.5	1,497
Treated	879.9	853.0	−3.1	377
Control	828.1	1,060.4	+28.1	1,120
Poor (baseline class)				
Treated	590.3	387.4	−34.4	77
Control	1,037.1	1,111.0	+7.1	269
Vulnerable (baseline class)				
Treated	1,020.2	1,101.5	+8.0	155
Control	939.3	990.3	+5.4	441
Middle-Upper (baseline class)				
Treated	926.9	899.7	−2.9	145
Control	559.9	1,098.7	+96.2	410

Notes: Baseline (2022) welfare classes as in Table D6; population-weighted, *Panel A* reports the two ingredients of the price shock: the national 2022–2024 change in household unit values by food item (pooling treated and control; p5–p95 trimmed), and each baseline welfare class's 2022 food budget share (food spending as a share of total spending), with the poor spending the largest share. Unit values are spending divided by quantity, so a measured rise can partly reflect a shift toward cheaper items or qualities rather than pure price inflation; we read them as indicative of the direction and rough magnitude of the shock. *Panel B:* value of own-consumed livestock, byproducts, and crops at aimag-median sold-unit prices, per household-year (2024 prices); classes held fixed, so each cell tracks the same households ("N" unweighted).

E Distributional Methods and Quantile Effects

E.1 FGT Poverty Measures

Table E1 reports the Foster et al. (1984) family of poverty measures: the headcount ratio (FGT_0, the share below the line), the poverty gap (FGT_1, the average proportional shortfall), and the squared poverty gap or severity (FGT_2, which weights deeper shortfalls more heavily), all population-weighted and computed against the 2024 poverty line.

Table E1: FGT Poverty Measures

	FGT_0 (Headcount)	FGT_1 (Gap)	FGT_2 (Severity)
Panel A: Difference-in-differences			
Treated × Post	0.152***	0.044***	0.014**
	(0.054)	(0.014)	(0.006)
Panel B: Triple difference			
Treated × Post	0.070	0.030***	0.011**
	(0.045)	(0.011)	(0.005)
Treated × Post × Herder	0.206**	0.036	0.009
	(0.084)	(0.024)	(0.011)
Controls × Post	Yes	Yes	Yes
Household + Year FE	Yes	Yes	Yes
Observations	4,254	4,254	4,254
Control mean, 2022	0.335	0.074	0.024
Herders	0.449	0.105	0.036
Non-herders	0.258	0.053	0.017

Notes: FGT_0 = headcount ratio, FGT_1 = poverty gap, FGT_2 = squared poverty gap (severity). Panel A reports the pooled difference-in-differences; Panel B the triple difference, where Treated × Post is the non-herder effect and Treated × Post × Herder the herder differential (the herder effect is their sum). All use the preferred covariate-adjusted TWFE specification (household and year fixed effects, baseline covariates and region interacted with post). The lower panel reports the population-weighted 2022 control-group mean of each measure, overall and split by baseline herder status. SEs clustered at baseline soum in parentheses; population weights. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

E.2 Interpreting the RIF Quantile Estimates

This appendix sets out how to read the unconditional quantile effects in Tables 3 and E2. The estimates come from recentered influence function (RIF) regressions, which convert a population-level distributional statistic into a household-level dependent variable whose OLS coefficients recover the marginal effect of covariates on that statistic (Firpo et al., 2009).

In Tables 3 and E2, the RIF is computed each year on the full national distribution (all households pooled, population-weighted); only the regression sample is restricted by herder status. Coefficients in herder/non-herder sub-panels therefore measure each subgroup's contribution to the shift in the national statistic, not within-group inequality or within-group quantiles.

E.3 Unconditional Quantile Effects on Food Consumption

Table E2: Unconditional Quantile Treatment Effects on Log Food Consumption

	(1) Q15	(2) Q30	(3) Q50	(4) Q70	(5) Q90
Panel A: DD – All Households					
Treated × Post	-0.294***	-0.190**	-0.148**	-0.141	-0.362***
	(0.091)	(0.074)	(0.072)	(0.092)	(0.118)
Observations	4,254	4,254	4,254	4,254	4,254
Panel B: DD – Non-Herders					
Treated × Post	-0.296***	-0.294***	-0.157***	-0.100	-0.270***
	(0.093)	(0.074)	(0.055)	(0.064)	(0.069)
Observations	2,992	2,992	2,992	2,992	2,992
Panel C: DD – Herders					
Treated × Post	-0.303*	-0.153	-0.210	-0.271	-0.531**
	(0.172)	(0.132)	(0.143)	(0.183)	(0.244)
Observations	1,262	1,262	1,262	1,262	1,262
Control mean (2022, MNT/day)			7,738		

Notes: Unconditional quantile treatment effects on log food consumption per adult equivalent, estimated as in Table 3 (RIF of national food-consumption quantiles; main covariate-adjusted specification). Panels B and C are herder-status subsample regressions; non-herder effects are largest at Q15 and Q30 and reappear at Q90, and the lower-tail concentration, alongside the FGT gap results, indicates a poor-household rather than herder-specific food loss (Section 5.3). "Control mean (2022, MNT/day)" is the population-weighted 2022 control-group mean of daily food consumption PAE. SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.

F Robustness

This appendix collects the falsification tests and robustness checks behind Section 5.5, varying one dimension at a time on the main DD/TD outcomes.

F.1 Falsification Tests

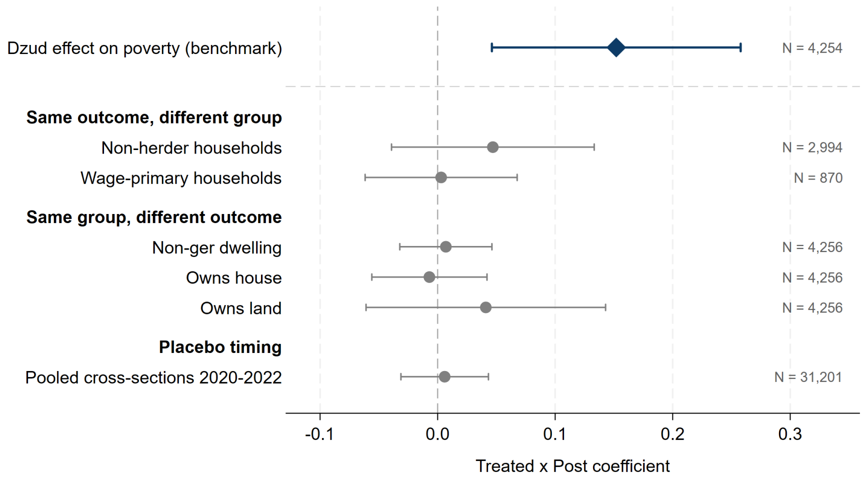


Figure F1: Falsification Tests: An Effect Only Where It Should Be

Notes: Each row is a separate difference-in-differences estimate with its 95 percent confidence interval; the vertical line marks zero. The top row is the paper's headline poverty effect. Below it, placebo estimates: groups without direct livestock exposure (non-herder households restricted to those without livestock income; wage-primary households), slow-moving placebo outcomes that a single winter shock should not move (non-ger dwelling; house and land ownership), and a placebo timed before the dzud (pooled cross-sections, 2020–2022, N=31,201). All outcomes are binary shares, so coefficients are comparable percentage-point effects. Standard errors clustered at the baseline soum; population weights. Full regression detail is available in the replication package.

F.2 Summary of Robustness

Table F1 collects the main DD and TD estimates under every robustness lever varied one at a time. The poverty difference-in-differences stays positive and significant (0.10–0.15) and the herder triple difference stays in the 0.08–0.26 range throughout. The triple difference attenuates to an insignificant 0.079 only when unweighted, so the herder effect is larger among bigger households—and because poverty is a population share, the population-weighted estimand is the policy-relevant one. Dropping the five Western aimags, which contain no treated soums under the P85 union and so enter only as controls, and re-estimating on the remaining 134 soums leaves the poverty DD essentially unchanged (0.152) and the herder triple difference if anything larger (0.206 to 0.232), each significant.

Table F1: Robustness of the Main Estimates

	Poverty	Log income	Log food
Panel A: Difference-in-differences (Treated × Post)			
Population weights (baseline)	0.152***	-0.113*	-0.210***
Household weights	0.123***	-0.112**	-0.186***
No weights	0.102***	-0.073	-0.187***
Exclude Q1 interviews	0.151**	-0.121*	-0.249***
Drop Western aimags	0.152***	-0.115*	-0.222***
Long-term stayers (5+ yr)	0.162***	-0.114*	-0.211***
Poverty line × 0.80	0.130***		
Poverty line × 1.20	0.139**		
Panel B: Triple difference (Treated × Post × Herder)			
Population weights (baseline)	0.206**	-0.151	-0.067
Household weights	0.141*	-0.249***	-0.074
No weights	0.079	-0.234***	-0.023
Exclude Q1 interviews	0.261***	-0.227**	-0.140
Drop Western aimags	0.232**	-0.130	-0.148
Long-term stayers (5+ yr)	0.213**	-0.152	-0.068
Poverty line × 0.80	0.095		
Poverty line × 1.20	0.211**		

Notes: Each row re-estimates the main specification varying one dimension; columns report the difference-in-differences (Panel A, Treated × Post) and the triple difference (Panel B, Treated × Post × Herder) for poverty, log income, and log food consumption per adult equivalent. "Population weights (baseline)" is the main specification; weighting rows vary the survey weights; "Exclude Q1 interviews" drops households interviewed in 2024 Q1; "Drop Western aimags" excludes region 1 (no treated soums); "Long-term stayers" restricts to heads resident 5+ years at baseline; the poverty-line rows rescale the threshold (poverty outcome only). Ulaanbaatar-as-control uses a different sample, reported separately (Table F6). All specifications use two-way (household, year) fixed effects with baseline covariates and region interacted with post. SEs clustered at baseline soum; population weights unless noted. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

F.3 Alternative Estimators

Table F2: Alternative Estimators: DD and TD ATT Estimates

	Poverty				Tot. Cons. (Log)				Income (Log)			
	OLS	TWFE	IPW	DR-DiD	OLS	TWFE	IPW	DR-DiD	OLS	TWFE	IPW	DR-DiD
Panel A: Difference-in-differences (ATT)												
ATT	0.126**	0.152***	0.121**	0.079***	-0.062	-0.141***	-0.064	-0.056	-0.085	-0.113*	-0.081	-0.076
	(0.049)	(0.054)	(0.048)	(0.023)	(0.051)	(0.048)	(0.050)	(0.035)	(0.053)	(0.060)	(0.055)	(0.049)
Panel B: Triple difference ($\text{ATT}_{\text{herder}} - \text{ATT}_{\text{non-herder}}$)												
DATT	0.201**	0.206**	0.189**		-0.110	-0.131*	-0.098		-0.123	-0.151	-0.117	
	(0.087)	(0.084)	(0.086)		(0.083)	(0.078)	(0.082)		(0.103)	(0.101)	(0.104)	
Observations	4,256	4,256	4,256	4,256	4,256	4,256	4,256	4,256	4,251	4,251	4,251	4,251
Control mean, 2022	0.335	0.335	0.335	0.335	22,846	22,846	22,846	22,846	9,199,219	9,199,219	9,199,219	9,199,219

Notes: OLS reports unconditional DD estimates. TWFE uses two-way fixed effects (household and year FE) with baseline covariates and region indicators interacted with post, the preferred specification from Table 2. Panel A reports the difference-in-differences ATT; Panel B the triple-difference DATT (the herder minus non-herder ATT). IPW (Abadie, 2005) and DR-DD (Sant'Anna and Zhao, 2020) condition on 15 baseline covariates for the DD, and IPW on 14 for the TD (dropping herder, the moderator). DR-DD is omitted from Panel B: assembled from two covariate-adjusted subsample DDs, it estimates the covariate-adjusted differential (CDATT) rather than the raw DATT the paper targets, and is not comparable to the other columns (Section 5.5). "Control mean, 2022" is the population-weighted 2022 control-group mean of the dependent variable. R-squared for fixed-effects specifications is within R-squared. SEs clustered at baseline soum in parentheses; population weights. * p<0.10, ** p<0.05, *** p<0.01.

F.4 Coefficient Stability (Oster 2019)

Table F3: Oster (2019) Coefficient Stability

	Short (no ctrl)	Long (with ctrl)	R-sq (short)	R-sq (long)	δ	N
Panel A: DD (Treated × Post)						
Poverty	0.126**	0.151***	0.011	0.077	-17.1	2,128
Log total consumption	-0.062	-0.140***	0.004	0.072	-5.7	2,128
Log food consumption	-0.129**	-0.210***	0.010	0.058	-7.2	2,128
Log income	-0.086	-0.116*	0.003	0.090	-12.5	2,123
Panel B: TD (Treated × Post × Herder)						
Poverty	0.201**	0.199**	0.020	0.083	234.4	2,128
Log total consumption	-0.110	-0.121	0.015	0.075	-28.1	2,128
Log food consumption	-0.067	-0.076	0.011	0.059	-22.3	2,128
Log income	-0.120	-0.158	0.024	0.092	-10.3	2,123

Notes: Coefficients from first-differenced regressions (equivalent to TWFE in a 2-period balanced panel). The panels report the main DD coefficient (Treated) and the TD coefficient (Treated × Herder). "Short" = without controls; "Long" = with the full baseline covariate set (14 for the DD; 13 for the TD, dropping herder), interview quarter, and region indicators. δ computed with $R_{\max} = \min(1,\ 1.3 \times R^2_{\text{long}})$ following Oster (2019); $\delta > 1$: robust to proportional selection on unobservables; $\delta < 0$: controls strengthen the coefficient. Full sample excl. Ulaanbaatar. SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.

F.5 Alternative Treatment Definitions

Table F4 reports the main DD and TD across alternative treatment definitions: climate-based definitions at varying severity thresholds (Panel A) and, as a descriptive benchmark, definitions built from soum-level shares of self-reported livestock losses (Panel B), which are endogenous to household behavior and reporting rather than a causal design. Two patterns deserve stating plainly. First, estimates attenuate as the threshold loosens: reclassifying moderately exposed soums as treated dilutes the treated–control contrast, so the P75 definitions are smaller and less precise than the P85 baseline. This gradient is what a genuine severity effect implies; identical estimates at every cutoff would be the suspicious pattern. Second, the herder triple difference is the more stable parameter, remaining significant at the 80th percentile and across the three-hazard snowfall unions at every threshold; the drought unions show the same pattern at the 10 percent level at P85 and P80, but not at P75.

Table F4: Robustness to Alternative Treatment Definitions

	Difference-in-differences			Triple difference		
	Poverty	Log income	Log food	Poverty	Log income	Log food
Panel A: Climate-based definitions						
Temp or Snow P85 [n=530]	0.152***	-0.113*	-0.210***	0.206**	-0.151	-0.067
	(0.054)	(0.060)	(0.075)	(0.084)	(0.101)	(0.120)
Temp or Snow P80 [n=616]	0.144***	-0.063	-0.179***	0.177**	-0.081	0.019
	(0.054)	(0.061)	(0.068)	(0.078)	(0.097)	(0.106)
Temp or Snow P75 [n=752]	0.094	-0.092	-0.168**	0.124	-0.063	0.032
	(0.059)	(0.070)	(0.076)	(0.077)	(0.091)	(0.099)
T/S/Snowfall P85 [n=876]	0.099***	-0.062	-0.161***	0.185**	-0.095	-0.161
	(0.036)	(0.042)	(0.050)	(0.073)	(0.090)	(0.106)
T/S/Snowfall P80 [n=1072]	0.081**	-0.043	-0.139***	0.167**	-0.021	-0.031
	(0.036)	(0.042)	(0.046)	(0.071)	(0.089)	(0.100)
T/S/Snowfall P75 [n=1221]	0.055	-0.054	-0.126**	0.153**	-0.029	-0.026
	(0.038)	(0.045)	(0.049)	(0.074)	(0.091)	(0.101)
Temp/Snow/Drought P85 [n=855]	0.098**	-0.118**	-0.139**	0.130*	-0.098	-0.088
	(0.043)	(0.047)	(0.062)	(0.074)	(0.083)	(0.098)
Temp/Snow/Drought P80 [n=960]	0.096**	-0.071	-0.121**	0.126*	-0.041	-0.024
	(0.043)	(0.048)	(0.059)	(0.072)	(0.083)	(0.094)
Temp/Snow/Drought P75 [n=1208]	0.057	-0.087	-0.084	0.120	-0.028	-0.017
	(0.052)	(0.059)	(0.072)	(0.073)	(0.085)	(0.099)
Panel B: Self-reported exposure						
Self-report soum >50% [n=492]	0.061	-0.127**	-0.176***	0.188**	-0.034	-0.109
	(0.046)	(0.051)	(0.049)	(0.077)	(0.101)	(0.087)
Self-report soum ≥75% [n=307]	0.118*	-0.107*	-0.197***	0.195**	-0.030	-0.054
	(0.061)	(0.056)	(0.067)	(0.093)	(0.097)	(0.109)

Notes: Each row is a separate treatment definition; the left block reports the difference-in-differences (Treated × Post) and the right block the herder triple difference (Treated × Post × Herder) for poverty, log income, and log food consumption per adult equivalent, under the preferred covariate-adjusted TWFE specification. n is the number of treated panel households; "Temp or Snow P85" is the paper's main definition. Panel B definitions are based on households' self-reported livestock losses aggregated to the soum; unlike the climate-based definitions in Panel A, they are endogenous to household behavior and reporting, and are shown as a descriptive benchmark rather than a causal design. Specification, sample, and weights as in Table 2; SEs clustered at baseline soum in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The self-reported definitions in Panel B point the same way, with the pattern one would expect from an exposure measure that embeds behavior. Defining treatment by the share of households in a soum reporting dzud livestock losses yields herder triple differences comparable to the climate-based rows (0.188 and 0.195 at the majority and 75 percent cuts, both significant at the 5 percent level) and food-consumption declines of similar magnitude (−0.176 and −0.197, against −0.210 under the main climate definition). The poverty DD, by contrast, is smaller and imprecise at the majority cut (0.061), consistent with treatment status that partly reflects households' own coping and reporting rather than exposure alone. We therefore read Panel B as a consistency check on the climate-based designs, not as an independent causal estimate.

A complementary check drops the middle of the severity distribution rather than varying the threshold. Table F5 keeps the treated group fixed and restricts the control group to the least-affected soums, first below median severity, then the bottom tercile. The poverty DD and the herder TD are essentially unchanged, so the headline is not generated by comparing severely against moderately affected soums; the same contrast emerges against soums that were barely touched.

Table F5: Extremes-Only Comparison: Dropping Moderately Affected Controls

	Full	Controls below median severity	Controls bottom tercile
	(1)	(2)	(3)
Poverty (DD)	0.152***	0.152***	0.145**
	(0.054)	(0.053)	(0.057)
Poverty (TD, herder)	0.206**	0.199**	0.199**
	(0.084)	(0.079)	(0.082)
Observations	4,254	3,540	2,866
Soums	175	134	105
Control soums	129	88	59

Notes: Column 1 replicates the main covariate-adjusted specification. Columns 2–3 keep all treated soums and progressively restrict the control group to the least-affected soums, ranked by the maximum of the temperature and snow-cover percentile ranks across the 175 analysis soums: below the median (col. 2) and in the bottom tercile (col. 3). Specification and weights as in Table 2; SEs clustered at baseline soum in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

F.6 Ulaanbaatar-as-Control Robustness

As a symmetric counterpart to the West-drop check, we add the nine UB districts back as controls (overriding their P85 classification) and re-estimate with the same specification, dropping the region-by-post interaction since UB falls outside the four-region scheme.

The poverty DD attenuates from 0.152 to 0.109 (Table F6, Panel A), as expected, since UB households recovered with the post-pandemic urban economy and pull the control-side change upward; the TD, by contrast, is essentially unchanged (0.206 to 0.220), because it identifies off the within-soum herder differential rather than the urban–rural composition. Panel B extends the DD to income and consumption.

Table F6: UB-as-Control: Poverty DD/TD and DD across Outcomes (Controls Included)

Panel A: Poverty DD and TD

	DD (poverty)		TD (poverty)	
	(1) Non-UB	(2) With UB	(3) Non-UB	(4) With UB
Treated × Post (DD)	0.152***	0.109**	0.070	0.029
	(0.054)	(0.048)	(0.045)	(0.040)
Treated × Post × Herder (TD)			0.206**	0.220**
			(0.084)	(0.090)
Soum clusters	175	184	175	184
Observations	4,254	5,006	4,254	5,006
R-squared	0.724	0.732	0.726	0.734

Panel B: DD across outcomes

	Poverty		Log Income		Log Total Cons.		Log Food Cons.	
	(1) Non-UB	(2) With UB	(3) Non-UB	(4) With UB	(5) Non-UB	(6) With UB	(7) Non-UB	(8) With UB
Treated × Post	0.152***	0.109**	-0.113*	-0.079*	-0.141***	-0.060	-0.210***	-0.127**
	(0.054)	(0.048)	(0.060)	(0.047)	(0.048)	(0.050)	(0.075)	(0.050)
Soum clusters	175	184	175	184	175	184	175	184
Observations	4,254	5,006	4,246	4,996	4,254	5,006	4,254	5,006
R-squared	0.724	0.732	0.770	0.783	0.813	0.855	0.605	0.643

Notes: Panel A reports the poverty DD (cols. 1–2) and the TD with baseline herder status (cols. 3–4); Panel B the DD (Treated × Post) on poverty, log income, log total and log food consumption. "Non-UB" is the main panel's covariate-complete estimation sample (175 soums, 4,254 household-years); "With UB" adds 9 UB districts forced to control (184 soums, 5,006 household-years; region × post dropped, as UB has no region code), with UB baseline covariates from the same raw HSES modules as non-UB. TWFE (household, year) plus baseline covariates × post; treatment Temp or Snow P85; SEs clustered at baseline soum; population weights. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

G Livestock Insurance and Dzud Impact

Table G1: Effect of Dzud on Herders, by Baseline Insurance Status

	Poverty		Food Cons.		Income		Triple Interaction		
	No Ins.	Insured	No Ins.	Insured	No Ins.	Insured	Poverty	Food	Income
Treated × Post	0.407**	0.123	-0.320*	-0.059	-0.351***	-0.092	0.362**	-0.346**	-0.282**
	(0.167)	(0.110)	(0.177)	(0.177)	(0.132)	(0.107)	(0.148)	(0.169)	(0.140)
Treated × Post × Insured							-0.135	0.128	0.078
							(0.163)	(0.209)	(0.173)
Observations	694	568	694	568	694	566	1,262	1,262	1,260
Clusters	118	91	118	91	118	91	156	156	156
R-squared	0.751	0.725	0.623	0.640	0.800	0.795	0.728	0.580	0.788
Insurance	Uninsured	Insured	Uninsured	Insured	Uninsured	Insured			
Sample	Herders	Herders	Herders	Herders	Herders	Herders	Herders	Herders	Herders

Notes: Herder subsample. Outcomes: poverty (0/1), log food and log income PAE. Cols. 1–6 stratify by baseline insurance status (household paid livestock insurance *or* tax); cols. 7–9 pool with a Treated×Post×Insured interaction. The headline estimate is the pooled poverty interaction: −0.135, with a 95 percent confidence interval of roughly [−0.46, 0.19], negative but not statistically significant. TWFE (household, year) plus baseline covariates and region × post. SEs clustered at baseline soum; population weights. * p<0.10, ** p<0.05, *** p<0.01.